Duhamel Neural Operator for Learning Phase-Field Dynamics

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Abstract

Phase-field simulators are used to predict how polarization fields evolve, a task that has been challenging due to their high computational cost. Machine learning surrogates have quicker inference times, but tend to struggle in approximating the amplification of incubation noise. We introduce the Duhamel Neural Operator (Duhamel-NO), a neural operator architecture consisting of a Fourier state-space model and a next-step U-Shaped Neural Operator, whose update is built on Duhamel's principle applied to the Time Dependent Ginzburg–Landau (TDGL) theory. Duhamel-NO reproduces the reference solver's (FerroX) trajectories up to three orders of magnitude faster while exhibiting the most physically consistent predictions and significantly lower error than the best of seven tuned baselines, achieving a near-perfect correlation zero-shot generalization on unseen materials, initializations, and free energy landscapes. Additionally, we construct an open benchmark for ferroelectric phase-field dynamics. Data & Code: `github.com/matthew-ju/Duhamel-Neural-Operator`

1 Introduction

Ferroelectric materials store information in the direction of their spontaneous polarization. That is, because the positive and negative ions are slightly displaced, each crystal cell of such a material carries a small electric dipole. These dipoles settle into one of two stable states, pointing up or down, because these states minimize their potential energy (Figure 1b). A voltage can switch the direction, and the material keeps the new state after the field is removed. Hafnia-based ferroelectric films, made of hafnium oxide (HfO_2), have been increasingly adopted for memory and logic devices because of their higher dielectric constants, which allow the gate insulators to be physically thicker while maintaining the same electrical capacitance (Böscke et al., 2011; Schroeder et al., 2022). This reduces quantum mechanical tunneling leakage currents, electrical leakage and chip heating, which allows devices to overcome the physical scaling limits of Moore's Law. This positions hafnia as a suitable substitute for, or addition to, silicon-based designs.

Importantly, the switching of the dipoles' directions is neither uniform nor instantaneous. The polarization is split into domains, which are regions in which the polarization points up or down, separated by thin domain walls. As such, the domain pattern determines how a film behaves. It sets the voltage needed to switch the film, the charge that the switch delivers, and how much two nominally identical devices differ. Predicting how domains form is thus a necessary task in device design.

The standard model of domain formation is the phase-field method (Chen, 2002). This tool assigns the out-of-plane polarization $P_z(\mathbf{r}, t)$ a free energy, which has a double-well shape in P_z, with one minimum for each

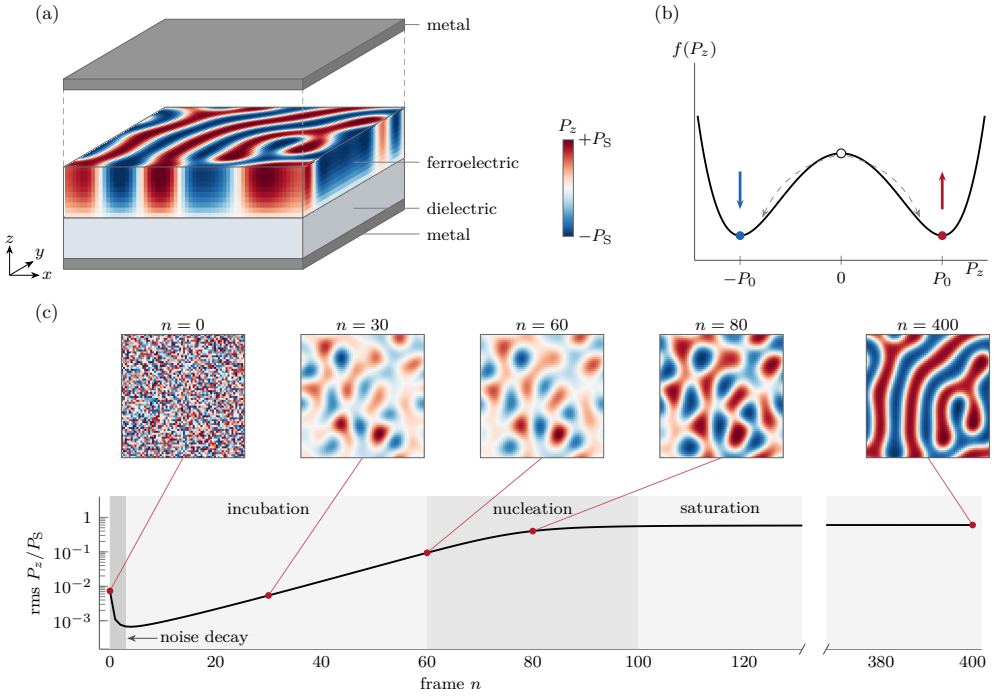


Figure 1: The problem. (a) The simulated device. A ferroelectric film sits on a dielectric layer between two metal electrodes (the top electrode is lifted for clarity). The faces of the film show P_z at the end of a FerroX simulation. Red domains point up, and blue domains point down. (b) The Landau energy of the same material. The unpolarized state $P_z = 0$ is unstable, and the minima $\pm P_0$ are the up and down states. (c) One trajectory from random noise. The snapshots show the top layer of the film, each on its own colour scale. In the first frames (dark band), the grid-scale noise decays quickly, and this is where truncated neural operators fail (Section 5). The pattern is set during incubation (compare $n = 30$ and $n = 60$). The stage boundaries differ between materials. In D1, nucleation starts anywhere from about frame 25 to frame 100. The task is to predict every frame from $n = 0$ and the material (α, β, γ).

direction (Figure 1b). The field flows downhill in this energy, following the time-dependent Ginzburg–Landau (TDGL) equation, coupled to the electrostatics of the device. Graphics processing unit (GPU) solvers such as FerroX (Kumar et al., 2023) solve these equations accurately, but a 3D trajectory is computationally expensive. This is consequential, as a study that samples n values of each of the three Landau coefficients needs n^3 trajectories, which quickly adds up to hundreds, if not thousands, of GPU-hours. A learned surrogate would make such studies cheap.

Current machine learning surrogates struggle with this problem because a trajectory starts from small random noise (about 0.5–2% of the saturated polarization P_S) and then passes through three stages (Figure 1c). 1) Incubation: the field stays small and evolves linearly, but a band of Fourier modes grows exponentially, and all other modes decay. 2) Nucleation: the growing modes become large and form domains. 3) Saturation: the domains reach $\pm P_S$ and slowly coarsen.

The final domain pattern is set by which noise modes are most prominent during incubation. As such, a surrogate that is wrong in the first frames tends not to recover the right pattern later. We identify this stage as the largest weakness of neural surrogates. They have to learn a response that grows on some modes and decays quickly on others, on a field that is about a thousand times smaller than the saturated one.

We show that FNOs are inherently unfit for the first few frames, in which the dynamics are linear. At this stage, each Fourier mode of the noise is multiplied by its own factor, which is large on a few growing modes and different on every decaying one. But because an FNO only processes a fixed set of low Fourier modes, all modes above this cut-off pass through the same pointwise layers, so they are all multiplied by one shared factor. Therefore, the FNO does not follow the different decay rates of the high modes, and this puts a ceiling on the pattern correlation it can reach, regardless of how well it is trained or implemented. We call this

ceiling the truncation cap. On our materials, the cap at frame 1 is 0.853, and a trained FNO reaches 0.826, just below it. The cap only binds in the first few frames, while most of the noise is still in the high modes, but this is the stage that decides the pattern.

These observations suggest two desiderata for a surrogate, namely fidelity in the linear stage, where the pattern is decided, and transfer to new materials without retraining. We address both by starting from the fundamental basis of the problem, the TDGL equation. The right-hand side of the TDGL equation splits exactly into a linear part and a nonlinear part. Duhamel's principle writes the subsequent state as the sum of two terms: 1) the linear dynamics, which we prove is a diagonal Fourier state-space model (SSM) computed in closed form from simulation data, and 2) an integral of the nonlinear force, which a neural operator learns as the correction that stops the exponential growth and saturates the field. We call the result the Duhamel Neural Operator (Duhamel-NO). Its Fourier SSM acts on every Fourier mode, so the cap does not apply, and it reaches a frame-1 correlation of 0.997.

The same split tells us how to condition the model on the material. The well depth α enters only the linear part, and the well shape, set by β and γ, enters only the nonlinear part. We train one model on a set of materials and test it without retraining on held-out materials, on materials outside the training range, and on wells of a different shape. Every test trajectory starts from noise that the model has never seen.

In sum, we contribute the Duhamel Neural Operator, a novel neural operator whose update is Duhamel's principle for the TDGL equation. Its architecture consists of a diagonal Fourier SSM that advances every Fourier mode by its exact linear factor, and a network that learns the nonlinear correction (Section 4). Additionally:

- We prove that an FNO, whose linear response is constant above its mode cut-off, cannot exceed a pattern correlation set by the linear dynamics alone, while Duhamel-NO has no such cap (Section 5).
- Without retraining and from unseen noise, Duhamel-NO reaches a pattern correlation of 0.982 at frame 400 on 256 held-out materials and 0.978 on 300 materials partly outside the training range. Its error is 23–26% lower than that of the best of seven tuned baselines at frame 400, and 2.61× lower at frame 1.
- Its predicted fields give better nucleation times, saturation polarization, electrostatic fields, free energy and other physical quantities than every baseline, at 1814× the throughput of the solver (Section 7).
- We release what is, to our knowledge, the first public benchmark for ferroelectric phase-field dynamics, with 1001 3D trajectories over a Latin hypercube of Landau coefficients, 300 trajectories with independently drawn coefficients, partly outside that range (D2), and 174 voltage-driven hysteresis loops (Section F).

2 Related Work

Phase-field models are standard for ferroelectric domain formation (Chen, 2002), especially when used in a GPU solver such as FerroX (Kumar et al., 2023). However, generating 3D simulations is computationally expensive and time-consuming. The goal of machine learning surrogates is to provide an alternative to these traditional frameworks, simulating domain dynamics at high accuracy with a fraction of the cost and time. Since the seminal work of Montes de Oca Zapiain et al. (2021), surrogates for phase-field systems follow the same recipe with convolutional networks, neural operators or DeepONets (Oommen et al., 2022; Li et al., 2023; Bonneville et al., 2025a;b), and some condition on material parameters (Fantasia et al., 2026; Hamdi & Lejeune, 2025). Similarly for ferroelectrics, it is standard to roll the polarization forward with a learned model, such as an autoregressive network (Alhada-Lahbabi et al., 2024) or a long short-term memory (LSTM) network over principal components (Alhada-Lahbabi et al., 2023). While these are highly effective methods that lead to state-of-the-art accuracy and speed-up, such surrogates are trained and tested at fixed coefficients and on initial noise that was seen during training. This sort of characteristic is known to inflate reported accuracy and is actually popular in scientific machine learning (Kapoor & Narayanan, 2023; McGreivy & Hakim, 2024). We therefore test only on unseen materials and unseen noise.

Most of these surrogates are built on general-purpose neural operators (Li et al., 2021; Kossaifi et al., 2024; Tran et al., 2023; Rahman et al., 2023; Raonić et al., 2023) or U-Nets (Gupta & Brandstetter, 2023; Ohana et al., 2024). They are trained to map one frame to the next and are applied autoregressively. Much

recent work targets the stability of long rollouts, for example by training on the model's own predictions (Brandstetter et al., 2022) or by refining each prediction with a diffusion-like process (Lippe et al., 2023), and foundation models pretrained on many equations are now available (Herde et al., 2024). All of these methods learn the whole update from data. In noise-seeded dynamics this is hardest in the first frames, where the field is small and the update is linear but different on every Fourier mode. We show that truncated neural operators cannot represent this linear stage exactly (Section 5).

A different line of work avoids learning the stiff linear part. Exponential time differencing integrates the linear part of an equation exactly and approximates only the nonlinear term (Cox & Matthews, 2002; Kassam & Trefethen, 2005; Hochbruck & Ostermann, 2010). Our update is the exponential-Euler member of this family, with the nonlinear term replaced by a neural operator. Several neural operators also integrate a Fourier symbol exactly. Neural stochastic partial differential equations (SPDEs) use the mild (Duhamel) form of the equation (Salvi et al., 2022), and SSNO and SGNO integrate a learned, dissipative symbol (Zhang et al., 2025; Li et al., 2026). These models assume or enforce a linear part that decays. In TDGL, the linear part must grow on a band of modes, because this instability seeds the domain pattern. We therefore measure the linear operator from simulations instead of learning it. This also lets it couple the depth layers of a 3D film and be interpolated across materials.

The measured operator connects our model to two further areas. Deep state-space models (SSMs) became practical once the state transition was made diagonal (Gu et al., 2022a;b; Gupta et al., 2022; Orvieto et al., 2023), and they have recently been combined with neural operators (Buitrago Ruiz et al., 2025; Koren & Lanthaler, 2026). The linear part of TDGL is exactly a diagonal SSM over Fourier modes (Proposition 3). Unlike in deep SSMs, its transition is not learned by gradient descent. It is computed in closed form from pairs of snapshots, which is the exact dynamic mode decomposition (DMD) or Koopman construction (Schmid, 2010; Tu et al., 2014; Kutz et al., 2016). Linearizing about the uniform state is also the classical theory of spinodal decomposition (Cahn & Hilliard, 1958; Cahn, 1961; Allen & Cahn, 1979), which explains why a band of modes grows and why it sets the domain size.

3 Background and Problem Setup

This section defines the physical system and the learning task. It assumes no background in ferroelectrics.

The device. We simulate the device of Figure 1a. A ferroelectric film, 5 nm thick, sits on a 4 nm dielectric layer between two metal electrodes. The electrodes are connected to each other, so no voltage is applied. The device measures 32×32 nm in the plane and is periodic in both in-plane directions. The state of the device is the out-of-plane polarization P_z of the film. It is the electric dipole moment per unit volume, in $\mathrm{C/m^2}$, and its sign gives the direction of the dipoles. We discretize the film on a $64 \times 64 \times 10$ grid, with ten layers through its thickness.

The free energy. The phase-field model assigns the polarization field the free energy

$$F[P_z] = \int \Big[\underbrace{\tfrac{\alpha}{2} P_z^2 + \tfrac{G_{11}}{2} \|\nabla P_z\|^2}_{\text{quadratic}} + \underbrace{\tfrac{\beta}{4} P_z^4 + \tfrac{\gamma}{6} P_z^6}_{\text{higher order}} \Big] d\mathbf{r} + U_{\text{elec}}[P_z]. \tag{1}$$

It has three parts.

- The **Landau energy** $f(P_z) = \frac{\alpha}{2} P_z^2 + \frac{\beta}{4} P_z^4 + \frac{\gamma}{6} P_z^6$ is the energy density of a uniformly polarized cell (Figure 1b). The coefficient $\alpha < 0$ makes the unpolarized state $P_z = 0$ a maximum, so the cell prefers to polarize. It sets the depth of the double well. The coefficients β and γ set where the two minima lie and how steep the wells are. If $\beta < 0$, the quartic term alone would let the energy fall without bound, and the sextic term ($\gamma > 0$) keeps it bounded. Such wells are called first order. The three Landau coefficients (α, β, γ) define the material.
- The **gradient energy** $\frac{G_{11}}{2} \|\nabla P_z\|^2$ penalizes rapid changes of P_z in space. A domain wall therefore costs energy, and the gradient coefficient G_{11} sets its width.

- The **electrostatic energy** U_{elec} comes from the charge that the polarization creates. A polarized region carries opposite charges on its top and bottom faces. These charges create a field E_z that points against the polarization, called the *depolarization field*. The electrodes cancel part of it, but the dielectric layer keeps them from cancelling all of it. The remaining field raises the energy of a uniformly polarized film. Neighbouring up and down domains carry opposite charges, so a domain pattern lowers this energy. This is why the film forms domains instead of polarizing uniformly. U_{elec} is quadratic in P_z, and computing it requires solving a Poisson equation for the electric potential in the whole device.

The dynamics. The polarization relaxes by gradient flow of F:

$$\frac{\partial P_z}{\partial t} = -\Gamma \frac{\delta F}{\delta P_z}. \tag{2}$$

Here $\delta F/\delta P_z$ is the variational derivative of F. It is the continuum version of the gradient of F with respect to the value of P_z at each point. Equation (2) therefore moves P_z at every point downhill in energy, at a rate set by the mobility Γ. Equation (2) is the TDGL equation. It is a stiff, nonlinear partial differential equation in 3D, and the solver must solve the Poisson equation at every time step.

From noise to domains. Each trajectory starts from small random noise, 0.5–2% of the final amplitude. This models a film that has just entered the ferroelectric phase, where small fluctuations seed the domains. The trajectory then passes through the three stages of Figure 1c. Inside a domain, P_z settles at the *saturated polarization* P_{S}. P_{S} is somewhat smaller than the minimum of f, because the depolarization field pushes back. The domain pattern depends on the noise. Two runs of the same material from different noise give different patterns with similar domain sizes. A surrogate must therefore predict the pattern of each individual trajectory, not only its statistics.

The learning task. Given the initial noise P^0 and the material (α, β, γ), the model must predict the frames $P^1, \ldots, P^{400}$. It is applied autoregressively, one frame at a time, and it never sees a solver frame after P^0. The frame interval is chosen per material so that the 400 frames cover the whole process (Section 6.1). The test materials and the test noise are both unseen in training. We measure accuracy mainly by the pattern correlation between the predicted and the true frames (Section 6.2).

4 Method

4.1 The linear and nonlinear parts of TDGL

As labeled in Equation (1), we split the right-hand side into two parts by taking the variational derivative: 1) the quadratic energy, which is a linear term represented by $\mathcal{L}P_z$, and 2) the higher-order energy, which is a nonlinear term represented by $\mathcal{N}(P_z)$. That is,

$$\partial_t P_z = \mathcal{L}P_z + \mathcal{N}(P_z), \qquad \mathcal{L}P_z = -\Gamma\big(\alpha P_z - G_{11}\nabla^2 P_z - E_z[P_z]\big), \qquad \mathcal{N}(P_z) = -\Gamma\big(\beta P_z^3 + \gamma P_z^5\big). \tag{3}$$

where E_z is the depolarization field, which is linear in P_z. The linear term, $\mathcal{L}$, is unstable on a band of wavenumbers, so it amplifies the noise modes that seed the domain pattern. The nonlinear term, $\mathcal{N}$, is of order P_z^3, and so it is negligible while the field is small. However, $\mathcal{N}$ opposes further growth and saturates the field at P_{S} if the field grows. (For $\beta < 0$ the quartic term pushes the other way, and the sextic term provides the restoring force.) Importantly, note that α is only in $\mathcal{L}$ and β and γ is only in $\mathcal{N}$.

4.2 Duhamel's principle

TDGL is nonlinear, so it has no closed-form solution. Duhamel's principle still writes one frame of it exactly, as a linear part plus an integral of the nonlinear force.

Proposition 1 (Duhamel form of TDGL)**.** Let $P_z(t)$ solve Equation (3), and let $\tau > 0$ be the frame interval. Then

$$P_z(t+\tau) \;=\; \underbrace{e^{\tau\mathcal{L}}\,P_z(t)}_{\text{linear part}} \;+\; \underbrace{\int_0^{\tau} e^{(\tau-s)\mathcal{L}}\,\mathcal{N}\big(P_z(t+s)\big)\,ds}_{\text{nonlinear correction}}. \tag{4}$$

Proposition 1 is the classical variation-of-constants formula (Hochbruck & Ostermann, 2010). It is usually stated for a linear equation $\partial_t u = \mathcal{L}u + f(t)$ with a known source f. To prove it, multiply this equation by the integrating factor $e^{-s\mathcal{L}}$. The terms with $\mathcal{L}$ cancel, and we are left with $\frac{d}{ds}\left[e^{-s\mathcal{L}}u(t+s)\right] = e^{-s\mathcal{L}}f(t+s)$. Integrating from $s = 0$ to $s = \tau$ and multiplying by $e^{\tau\mathcal{L}}$ gives Equation (4) with f in place of $\mathcal{N}$. TDGL is not linear, because $\mathcal{N}(P_z)$ depends on the field itself. Along a solution, however, $\mathcal{N}(P_z(t))$ is simply a function of time, so it can play the role of f. This proves Proposition 1.

The two terms of Equation (4) have simple meanings. The first moves the current state forward with the linear dynamics alone. Here $e^{\tau\mathcal{L}}$ is the operator exponential, which multiplies each eigenmode of $\mathcal{L}$ by $e^{\tau\mu}$, where μ is the eigenvalue of that mode. The second term adds the nonlinear force. The force at time $t+s$ is then moved forward by the linear dynamics for the rest of the frame, $\tau - s$.

Equation (4) is exact, but it is not a solution formula. The integral needs the field $P_z(t+s)$ at every time inside the frame, and this field is what we want to predict. It is useful because it separates the two parts of Equation (3). The linear part is computed exactly by $e^{\tau\mathcal{L}}$, for any frame length τ. Only the integral has to be approximated, and it comes from $\mathcal{N}$ alone. During incubation the field is small, so $\mathcal{N} = O(P_z^3)$ is close to zero, and so is the integral. The first term then gives almost the whole step. A model built on Equation (4) is therefore nearly exact in the first frames. A model that learns the whole step must also learn the linear part, and Section 5 shows that truncated neural operators cannot.

4.3 The Duhamel-NO step

The integral in Equation (4) depends on the unknown path inside the frame. The next proposition replaces it by a single function of the current field, which a network can learn.

Proposition 2 (Effective force)**.** Let $\mathcal{L}$ be symmetric, let $\varphi_1(z) = (e^z - 1)/z$, and let Φ_τ map one frame to the next, $P_z(t+\tau) = \Phi_\tau(P_z(t))$. Then there is a unique field N_{eff}, which depends only on $P_z(t)$, such that

$$P_z(t+\tau) = e^{\tau\mathcal{L}}P_z(t) + \tau\varphi_1(\tau\mathcal{L})\,N_{\text{eff}}(P_z(t)), \qquad N_{\text{eff}} = \left[\tau\varphi_1(\tau\mathcal{L})\right]^{-1}\left(\Phi_\tau(P_z) - e^{\tau\mathcal{L}}P_z\right). \tag{5}$$

If $\mathcal{N}$ is constant during the frame, then $N_{\text{eff}} = \mathcal{N}(P_z(t))$.

We prove the three parts of Proposition 2 in turn. First, N_{eff} depends only on $P_z(t)$. TDGL is deterministic, and its right-hand side does not depend on time explicitly. The current field therefore fixes the whole path inside the frame, and with it $\Phi_\tau(P_z(t))$. Second, N_{eff} is well defined, because $\tau\varphi_1(\tau\mathcal{L})$ is invertible. $\mathcal{L}$ is symmetric (Section A.1), so its eigenvalues μ are real, and the eigenvalues $\tau\varphi_1(\tau\mu)$ of $\tau\varphi_1(\tau\mathcal{L})$ are positive for every real μ. Third, $\tau\varphi_1(\tau\mathcal{L})$ is the integral of the kernel in Equation (4). If $\mathcal{N}$ keeps a constant value N_0 during the frame, N_0 comes out of the integral:

$$\int_0^\tau e^{(\tau-s)\mathcal{L}}N_0\,ds = \Big(\int_0^\tau e^{(\tau-s)\mathcal{L}}ds\Big)N_0 = \tau\varphi_1(\tau\mathcal{L})\,N_0. \tag{6}$$

The last equality holds mode by mode. On a mode with eigenvalue μ, the integral of $e^{(\tau-s)\mu}$ over s is $(e^{\tau\mu} - 1)/\mu = \tau\varphi_1(\tau\mu)$. Substituting Equation (6) into Equation (4) and comparing with Equation (5) gives $N_{\text{eff}} = N_0$.

In words, N_{eff} is the constant force that gives the same next frame as the true, changing one. Setting $N_{\text{eff}} = \mathcal{N}(P_z(t))$ in Equation (5) gives the exponential-Euler scheme (Cox & Matthews, 2002; Hochbruck & Ostermann, 2010). In general, N_{eff} differs from $\mathcal{N}(P_z(t))$ by $O(\tau)$ on modes that are not stiff (Section A.2), and it is close to zero during incubation.

N_{eff} has no closed form, but it depends only on the current field and the material. We therefore learn it with a neural operator N_ψ, which receives the field and a short vector of material numbers $\mathbf{m}$. We write the step in material units (Section 4.5). The field is $v = P_z/P_{\text{S}}$, and time is measured in units of the linear growth time λ, so the frame interval is $\theta = \tau/\lambda$. We write $L = \lambda\mathcal{L}$ for the linear operator in these units, so that $\theta L = \tau\mathcal{L}$. Equation (5) then keeps its form:

$$v_{n+1} \;=\; e^{\theta L}\,v_n \;+\; \theta\,\varphi_1(\theta L)\,N_\psi(v_n;\,\mathbf{m}). \tag{7}$$

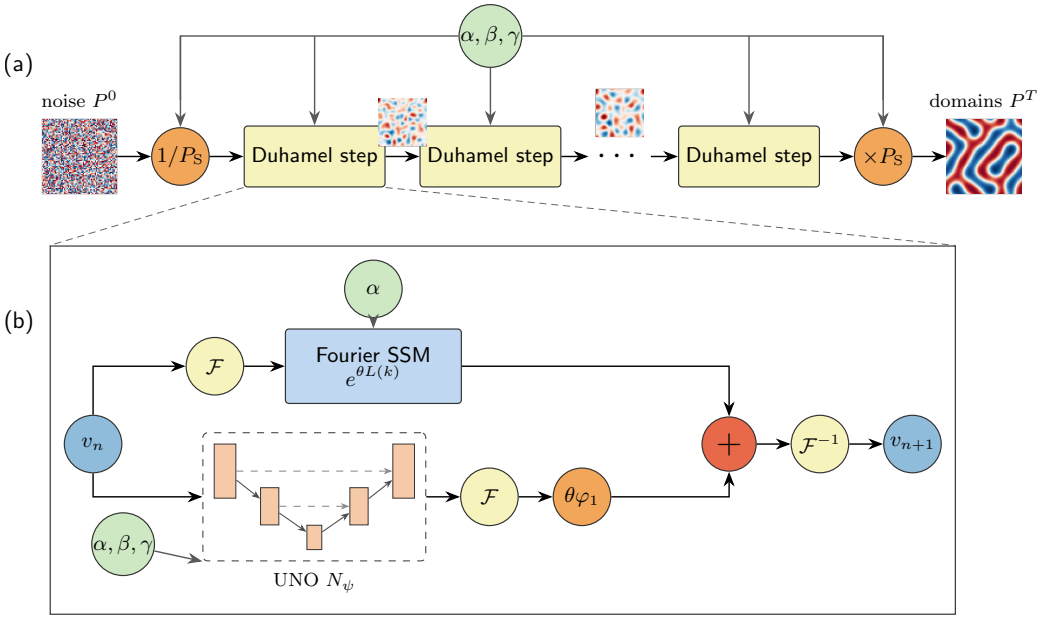


Figure 2: Duhamel-NO. (a) The initial noise P^0 is put in material units (divided by P_S) and advanced by repeated Duhamel steps. The material (α, β, γ) conditions every step. The thumbnails show incubation, nucleation and saturation. (b) One step (Equation (7)). The Fourier SSM $e^{\theta L(k)}$, set by α, multiplies every Fourier mode of v_n by its exact linear factor. The U-shaped neural operator (UNO) N_ψ gives the nonlinear correction. Its output is filtered by $\theta\varphi_1(\theta L)$ and added in Fourier space.

Here $e^{\theta L}$ and $\theta\varphi_1(\theta L)$ multiply each in-plane Fourier mode of v_n by a 10×10 matrix, $e^{\theta L(k)}$ and $\theta\varphi_1(\theta L(k))$ (Section 4.4). Each part of Equation (7) has one job (Figure 2b).

- The **Fourier SSM** $e^{\theta L}$ moves every Fourier mode by its exact linear factor. It decides which noise modes grow, and so it decides the domain pattern. It is measured from data, not learned (Section 4.4).
- The **filter** $\theta\varphi_1(\theta L)$ damps the network output on stiff modes, as the true integral does. It is computed from the same L.
- The **network** N_ψ learns the effective force. Its last layer starts at zero, so the untrained model is the exact linear solution.

The only approximation in Equation (7) is $N_\psi \approx N_{\text{eff}}$. Any error in the measured L is passed on to the network, which is trained so that the whole step matches the data.

4.4 The Fourier SSM

The Fourier SSM is the first term of Equation (7). We now show that it acts on each Fourier mode separately, explain which modes grow, and describe how we measure it.

$\mathcal{L}$ does not change under in-plane translations, so in-plane Fourier modes do not mix. Each mode evolves on its own. The electrostatics, however, couple the ten depth layers. For each in-plane wavevector k, the ten layer amplitudes $\hat{P}_z(k,t) \in \mathbb{C}^{10}$ therefore obey

$$\partial_t \hat{P}_z(k,t) = \mathcal{L}(k)\, \hat{P}_z(k,t), \qquad \mathcal{L}(k) = -\Gamma\big[(\alpha + G_{11}|k|^2)\, I + C(k)\big], \tag{8}$$

where $C(k)$ is the 10×10 layer-coupling matrix. It contains the finite differences across the film and the depolarization field. A mode grows when one of the eigenvalues of $\mathcal{L}(k)$ is positive. The terms of Equation (8) compete. $\alpha < 0$ drives growth on every mode. The gradient term $G_{11}|k|^2$ damps large $|k|$, and the depolarization field in $C(k)$ damps small $|k|$. Growth is therefore confined to a band of $|k|$, and this band sets the domain size. β and γ do not appear in Equation (8).

$\mathcal{L}(k)$ is real and symmetric, so it has real eigenvalues $\mu_i(k)$ and orthonormal eigenvectors $V(k)$. Each eigenvector is a pattern across the ten layers. Integrating Equation (8) over one frame gives the one-frame map

$$B(k) = e^{\tau \mathcal{L}(k)} = e^{\theta L(k)} = V(k)\,\mathrm{diag}\big(e^{\tau \mu_i(k)}\big)\,V(k)^\top. \tag{9}$$

So every frame, $B(k)$ multiplies each layer pattern by its own factor $e^{\tau \mu_i(k)}$. This is a diagonal state-space model.

Proposition 3 (The linear dynamics are a diagonal Fourier SSM)**.** Write $B(k) = V(k)\Lambda(k)V(k)^{-1}$. Then the linear dynamics are exactly the diagonal, linear, time-invariant state-space recurrence

$$h^0(k) = V(k)^{-1}\hat{P}_z{}^0(k), \qquad h^{t+1}(k) = \Lambda(k)\,h^t(k), \qquad \hat{P}_z{}^t(k) = V(k)\,h^t(k),$$

with $A = \Lambda$, $C = V$, $D = 0$, and the initial noise as an impulse input. This is the form of the diagonal SSMs S4D, DSS and LRU (Gu et al., 2022b; Gupta et al., 2022; Orvieto et al., 2023).

The proof is in Section A.3. Each in-plane mode is an independent copy of this system, so one step of the Fourier SSM costs one 10×10 matrix–vector product per mode.

We could compute $B(k)$ from Equation (8). However, the solver's discrete operators differ from the continuum ones, so we measure $B(k)$ from simulations instead. We do this in three steps (Algorithms 2 and 3 in Section B).

1. *Collect frame pairs.* While the field is small, $\mathcal{N}$ is negligible, and consecutive frames obey $\hat{P}_z{}^{t+1}(k) = B(k)\hat{P}_z{}^t(k)$. We use the pairs $t = 0, \ldots, 7$ of each training simulation.
2. *Fit one map per radial bin.* In Equation (8), k enters only through $|k|$. We therefore group the modes into 46 radial bins of $|k|$. For bin b, we place the frames $\hat{P}_z{}^t$ of all its modes in the columns of a matrix X, and the next frames $\hat{P}_z{}^{t+1}$ in the columns of Y. Least squares with a small ridge ϵ then gives B_b in closed form,
$$B_b = \arg\min_B \|Y - BX\|_F^2 = YX^{\mathsf{H}}\big(XX^{\mathsf{H}} + \epsilon I\big)^{-1}. \tag{10}$$
This is exact DMD (Schmid, 2010; Tu et al., 2014).
3. *Build the map of a new material.* We convert each map to its generator, $L_b = \theta^{-1}\log B_b$, and average the generators of the eight training materials closest in $\log|\alpha|$. We average generators rather than maps because the filter in Equation (7) needs L itself.

Some bins decay below round-off within one frame, so they cannot be measured. They get a learnable, strongly damped rate. On validation materials, the Fourier SSM alone reproduces frame 1 with a correlation of 0.997.

4.5 The correction network

The network N_ψ learns the effective force of Equation (5). Different materials saturate at different amplitudes and grow at different rates. We make them look alike by changing units, so that one network can serve them all.

We measure the field in units of P_{S}, $v = P_z/P_{\mathrm{S}}$, and time in units of the linear growth time λ. Both P_{S} and λ come from closed-form laws in (α, β, γ) that account for depolarization (Section C.3). In these units, the nonlinear force of every material has the same shape,

$$-(1-r)\,v^3 - r\,v^5, \tag{11}$$

up to a prefactor that depends on the material (Section A.4). The shape number r is the share of the restoring force that comes from the sextic term ($r > 1$ means $\beta < 0$). So β and γ enter the correction only through r. This change of units alone, before any training, reduces the coefficient of variation of the nucleation time across D1 materials from 0.74 to 0.054.

Algorithm 1 Rollout of Duhamel-NO from initial noise

Require: noise P^0; material (α, β, γ); number of frames T; trained N_ψ

$P_\mathrm{S}, \theta, \mathbf{m} \leftarrow$ material units (Section C.3); $L(k) \leftarrow$ Fourier SSM (Algorithm 3)

$E(k) \leftarrow e^{\theta L(k)}$, $\Phi(k) \leftarrow \theta\varphi_1(\theta L(k))$; $v \leftarrow P^0/P_\mathrm{S}$

for $n = 0, \dots, T-1$ **do**

  $h \leftarrow N_\psi(v; \mathbf{m})$ ▷ rms-normalized (Section 4.5)

  $v \leftarrow \mathrm{IFFT}\big(E\,\hat{v} + \Phi\,\hat{h}\big)$; $P^{n+1} \leftarrow P_\mathrm{S}\,v$ ▷ Equation (7); hats denote the in-plane FFT

return $P^1, \dots, P^T$

The network receives the material numbers $\mathbf{m} = (\log\theta, \log P_\mathrm{S}, r, \log\Pi_h)$, where Π_h is the grid spacing in domain-wall widths. The field still changes by about 10^3 between incubation and saturation. We therefore divide it by its root-mean-square (rms) amplitude s before it enters the network, and add a channel that holds $\log_{10} s$, so that the network knows how far the field is from saturation. The output is multiplied back by s.

N_ψ is a U-shaped neural operator (UNO) (Rahman et al., 2023), a Fourier neural operator arranged like a U-Net, with the ten depth layers as channels. It has 9.5M parameters, and its architecture is given in Section D. Like every Fourier neural operator, it keeps only the lowest Fourier modes (up to 16 per direction). This costs nothing during incubation, because the Fourier SSM in Equation (7) acts on every mode.

4.6 Training

Duhamel-NO maps frame P^n to frame P^{n+1}. At inference, it is applied to its own output (Algorithm 1). If we trained it on single steps, its small errors would compound at inference. We therefore train it on short rollouts. We sample windows of $k+1 = 5$ consecutive frames. Starting from the first frame of a window, we roll the model out for $k = 4$ steps and average the relative error of every step:

$$\mathcal{L} = \frac{1}{k}\sum_{q=1}^{k} \frac{\|\tilde{P}^q - P^q\|^2}{\|P^q\|^2}, \qquad \tilde{P}^0 = P^0, \quad \tilde{P}^q = \hat{\Phi}\big(\tilde{P}^{q-1}\big), \tag{12}$$

where $\hat{\Phi}$ is one step of Equation (7). Two choices in Equation (12) matter. First, we divide by $\|P^q\|^2$. The early fields are about 10^3 times smaller than the saturated ones, and a plain squared error would ignore them. Second, we backpropagate through all k steps. The pushforward trick (Brandstetter et al., 2022) instead stops the gradients through the model's own inputs.

The pattern is decided early, so 60% of the windows start in the first 120 frames, and the rest start anywhere. We also apply random exact symmetries (90° rotations, reflections and $P_z \to -P_z$) to every window. Optimizer settings are in Section D.

5 Why a Fourier SSM

In this section, we show that truncated neural operators cannot represent the first frames of a noise-seeded trajectory, however well they are trained, whereas a Fourier SSM can. The argument rests on two observations. While the field is small, the dynamics are linear and act on each Fourier mode separately, with a different factor for every mode. An FNO, by contrast, treats all modes above its cut-off identically. We first set up the notation and then state the two results.

We begin with the dynamics that a surrogate must reproduce during incubation, when the field is still small enough for the nonlinear term to be negligible. Let the initial noise ξ be white, so that its expected power is the same at every in-plane wavenumber k. While the field is small, one frame of TDGL is a linear map A that multiplies each Fourier mode k by a factor $g(k)$, the matrix $B(k)$ of Section 4.4. The true frame n is therefore $T_n = A^n\xi$, and its mode k is $g_n(k)\,\hat{\xi}(k)$ with $g_n(k) = g(k)^n$. These factors differ strongly between modes, since a narrow band of low wavenumbers grows while every other mode decays, each at its own rate.

An FNO, however, applies its spectral convolutions only to a fixed set $\mathcal{K}$ of low wavenumbers, which we call the retained mode set. We write $\mathcal{K}^c$ for the modes outside $\mathcal{K}$, which lie above the cut-off, and $|\mathcal{K}^c|$ for their number.

To compare a surrogate with the true dynamics, we need a measure of agreement that does not depend on a single realization of the noise, together with a description of how the surrogate responds to small inputs. Let a surrogate G map the noise ξ to a predicted frame $G(\xi)$. The population pattern correlation between the prediction and the truth is

$$\rho(G, T_n) = \frac{\mathbb{E}\langle \tilde{G}, \tilde{T}_n \rangle}{\sqrt{\mathbb{E}\|\tilde{G}\|^2 \, \mathbb{E}\|\tilde{T}_n\|^2}}, \tag{13}$$

where the tilde removes the spatial mean and the expectation is taken over the noise. The linear response of G is the part of $G(\xi)$ that is linear in ξ. When G is translation-equivariant and the noise is white, this part acts on each mode by a factor $m(k)$, so that its mode k is $m(k)\,\hat{\xi}(k)$.

With this notation, the question of how well G can match T_n reduces to a comparison of factors. Because T_n is linear in the noise, the part of $G(\xi)$ that is not linear in the noise is uncorrelated with T_n and can only lower ρ (Section A.6). The correlation is therefore governed by how closely $m(k)$ follows $g_n(k)$ across all modes. For small noise, $m(k)$ coincides with the Jacobian of G at the zero state, and for an FNO this Jacobian takes a particularly simple form.

Proposition 4 (FNO linearization)**.** For an FNO without positional channels, the Jacobian at any spatially uniform state is a Fourier multiplier $M(k)$, which equals one fixed matrix M_∞ for every $k \in \mathcal{K}^c$.

The reason is that each FNO layer combines a spectral convolution on the modes in $\mathcal{K}$ with a pointwise linear map and a pointwise nonlinearity. Outside $\mathcal{K}$ only the pointwise operations act, and a pointwise operation multiplies every Fourier mode by the same factor. Such a response cannot follow the distinct decay rates of the high modes, and the next proposition makes this limitation quantitative.

Proposition 5 (Truncation cap)**.** Let the noise be white, and let G be translation-equivariant with a linear response $m(k)$ that is constant on $\mathcal{K}^c$. Then

$$\rho(G, T_n) \;\le\; \rho_{\mathcal{K}}(n) = \Big(1 - \frac{|\mathcal{K}^c| \operatorname{Var}_{k \in \mathcal{K}^c}[g_n(k)]}{\sum_{k \neq 0} |g_n(k)|^2}\Big)^{1/2}. \tag{14}$$

By Proposition 4, the bound applies to an FNO without positional channels as the amplitude of the noise tends to zero.

The bound has a direct interpretation. Because the model must apply a single factor to every mode above the cut-off, its best choice is the average of g_n over those modes, and any departure of the true factors from this average is power that the prediction cannot reproduce. The cap therefore reflects the fraction of the true frame that lies in this irreducible residual. We give the proofs in Section A.6, together with two further results that describe how the correlation behaves as the amplitude of the initial noise tends to zero.

To evaluate the cap, we take the factors $g_n(k)$ from the Fourier SSM measured on the D1 held-out materials (Section 6.1) and use the mode set of the FNO in our experiments, which retains modes up to 8 in each in-plane direction. The resulting cap is 0.853, 0.882 and 0.926 at frames 1–3 and rises to 0.996 by frame 10 (Figure 3). The bound is most restrictive at the first frame because white noise distributes its power equally across all wavenumbers, so that most of the initial power lies above the cut-off, where the true decay rates are most heterogeneous. These high modes decay rapidly, however, and within a few frames their contribution to the field becomes negligible. From that point onward the field is governed by the growing band at low wavenumbers, which lies inside the retained mode set, and the cap accordingly ceases to constrain the prediction.

Proposition 5 holds rigorously for any model whose linear response is constant above the cut-off, yet the baselines we train depart from this idealization in two respects (Section 6.3). Rms normalization rescales even a vanishingly small input to unit amplitude, so the network never operates in the linear regime that the proposition describes, while coordinate channels break the translation equivariance on which the proof

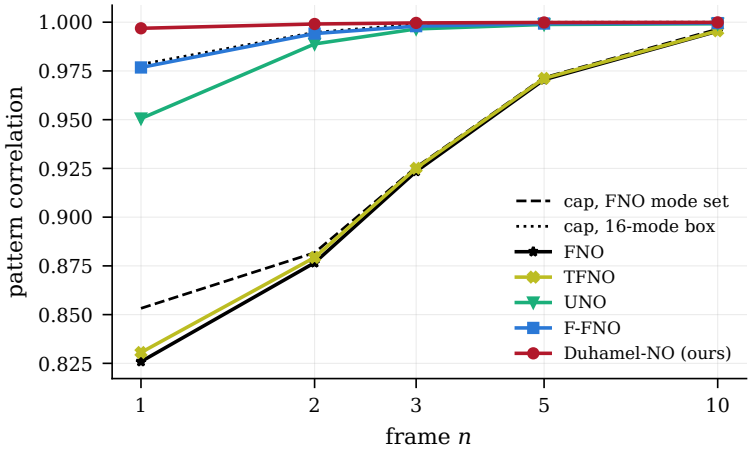


Figure 3: Truncation cap and trained models in the first ten frames (D1 held-out materials). Dashed and dotted lines show the best correlation that any model with the FNO's mode set, or with a 16-mode box, can reach (Proposition 5), computed from the measured Fourier SSM. Markers show trained models.

relies. For these models we therefore treat the cap as an empirical reference rather than a proven bound. Nor does the cap extend to architectures with a different mode set, such as F-FNO, or to architectures without spectral truncation, such as U-Nets and CNO. Duhamel-NO, by contrast, applies the Fourier SSM to every wavenumber, so its linear response may vary freely above any cut-off, and the bound places no restriction on it.

6 Experimental Setup

The analysis of Section 5 predicts that truncated neural operators fail mainly in the first frames of a noise-seeded trajectory, and testing this prediction requires data with three properties. Each trajectory must start from small random noise, so that the linear stage is resolved; the data must span many materials, so that generalization across materials can be measured; and every trajectory must start from its own noise, so that a model cannot score well by memorizing initial fields. The public datasets we examined do not meet these requirements. The Allen–Cahn and Cahn–Hilliard sets of Poseidon (Herde et al., 2024) and PDENNEval (Wei et al., 2024) start at 39–56% of their final amplitude, so their linear stage is largely absent, and PDEBench (Takamoto et al., 2022) reuses the same initial conditions across every parameter value of its 1D Burgers and advection sets. We therefore generate our own datasets with FerroX, as described below.

6.1 Simulations and datasets

Device. All data are FerroX simulations of the device of Section 3. The relative permittivity is $\varepsilon_{\mathrm{fe}}$=24 in the film and $\varepsilon_{\mathrm{de}}$=10 in the dielectric, with G_{11}=$10^{-9}\,\mathrm{V\,m^3/C}$ and Γ=$100\,(\Omega\,\mathrm{m})^{-1}$. Each run starts from small uniform random noise and is saved for 401 frames. The frame interval of each material is chosen so that the frames cover about 1.25× its time to saturation.

D1 (training and held-out materials). D1 contains 1001 materials drawn as a Latin hypercube (McKay et al., 1979) in three coordinates. These are the well depth, $|\alpha| \in [1.6, 4.0] \times 10^9\,\mathrm{V\,m/C}$ (sampled in $\log|\alpha|$), the saturated polarization, $P_{\mathrm{S}} \in [0.10, 0.43]\,\mathrm{C/m^2}$, and the well shape, $r \in [0, 1.4]$. The coefficients β and γ are solved from these coordinates. We train on 745 materials (682 for training and 63 for validation) and hold out 256. Of the held-out materials, 82 lie inside the sampled box and 174 at its edges, including every well with $r \geq 1.15$.

D2 (independent materials). D2 contains 300 further materials. Their α, β, γ are drawn independently, in five regions of 60 materials each:

- small P_S;
- large P_S, up to 1.8× the training maximum;
- large $|\alpha|$ with small P_S;
- first-order wells with r up to 2, beyond the training range;
- all of these together, with P_S up to 2.2× the training maximum.

D2 is used for testing only, without retraining. Each material in D1 and D2 is simulated once. Section C gives more detail and shows where the materials lie.

Unseen noise. FerroX generates the same initial field for the same seed, whatever the material. A model can therefore memorize shared seeds. With eight shared seeds, a model scores 0.99 on a seen seed but only 0.66–0.82 on an unseen one. We therefore give *every trajectory its own seed*. The largest correlation between a test and a training initial field is 0.023.

6.2 Metrics

We report two field metrics at frame t. The *pattern correlation* of a single frame is $\mathrm{corr}(A, B) = \sum_i (A_i - \bar{A})(B_i - \bar{B}) / [\sum_i (A_i - \bar{A})^2 \sum_i (B_i - \bar{B})^2]^{1/2}$, and the *relative error* is $\|A - B\|_2 / \|B\|_2$. Frame t=400 is the settled state. A low field error does not guarantee correct physics. We therefore also recompute, from each prediction, the quantities that a materials scientist would extract. These are the nucleation time, the saturation polarization, the domain size, the wall density, the distribution of P/P_S, the electrostatic potential, the depolarization field and the free energy. We fixed this list before evaluating any model. Every prediction is rolled out from the t=0 field, with no solver frame.

6.3 Baselines and training

We compare against seven neural operators and U-Nets. These are the Fourier neural operator (FNO) (Li et al., 2021), its tensorized (TFNO) (Kossaifi et al., 2024) and factorized (F-FNO) (Tran et al., 2023) variants, UNO, U-Net, ConvNeXt U-Net (CNextU-Net) (Gupta & Brandstetter, 2023; Ohana et al., 2024), and the convolutional neural operator (CNO) (Raonić et al., 2023). For a fair comparison, every baseline is trained with the same setup as Duhamel-NO. This means material units, the amplitude-aware input of Section 4.5, the loss of Equation (12) and the same augmentation. The hyperparameters of every model are tuned on the validation materials. This setup matters for the baselines. With standard training instead, no backbone exceeds a correlation of about 0.84, at a relative error of 1.0–2.9, which is no better than predicting zero (Table 4). We also tried generative models (PDE-Refiner (Lippe et al., 2023), ACDM diffusion (Kohl et al., 2023) and flow matching (Lipman et al., 2023)), a transformer (Transolver (Wu et al., 2024)) and a fine-tuned foundation model (Poseidon-T (Herde et al., 2024)). None of them learned the task (Table 6). Every model is trained for 20 000 steps, and the selected configuration is retrained with three seeds (one for the slowest models). Details are in Section D.

7 Results

7.1 Zero-shot generalization across materials and noise

Table 1 gives the main results. On the 256 held-out D1 materials, Duhamel-NO reaches a pattern correlation of 0.982 and a relative error of 0.129 at frame 400. Its error is 23% lower than that of the strongest baseline, F-FNO. On D2, which the model never saw and which is partly outside the training range, its error is 0.145. This is 26% lower than F-FNO. Three training seeds agree to within ±0.001 in correlation.

Duhamel-NO does not lose accuracy when it extrapolates. Its error is 0.120 on held-out materials inside the sampled box and 0.133 on those at its edges. Across the five D2 regions, its correlation stays between 0.974 and 0.985. It is 0.985 on materials whose P_S reaches 1.8× the training maximum, and 0.974 on first-order wells with r beyond the training range, which have a different free-energy landscape (Table 7). Figure 4 shows a typical held-out material and a first-order well.

Table 1: Accuracy on unseen materials and unseen noise (mean over training seeds; **best**, second). D1 has 256 held-out materials, and D2 has 300 independent materials, partly outside the training range. Correlation and relative error at frame 400, and relative error at frame 1. Poseidon-T is the best of the generative, transformer and pretrained models. Table 6 lists all models with standard deviations.

model	D1 corr.	D1 error	D2 error	frame-1 error
Duhamel-NO (ours)	**0.982**	**0.129**	**0.145**	**0.082**
F-FNO	0.974	0.167	0.195	0.215
U-Net	0.966	0.207	0.242	0.313
UNO	0.962	0.219	0.238	0.315
CNextU-Net	0.959	0.229	0.261	0.347
CNO	0.951	0.256	0.292	0.451
TFNO	0.949	0.272	0.265	0.564
FNO	0.947	0.270	0.268	0.568
Poseidon-T (pretrained)	0.553	2.917	2.570	4.342

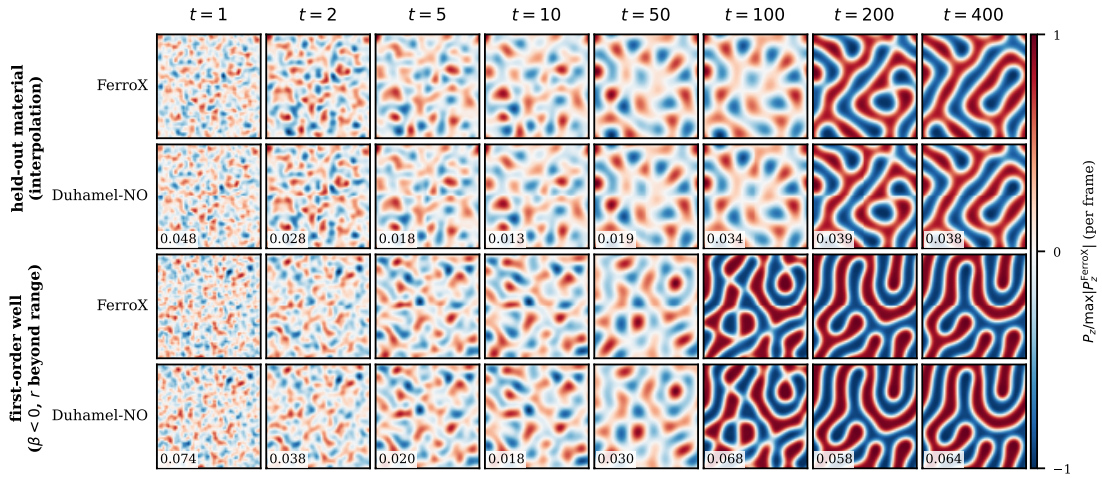


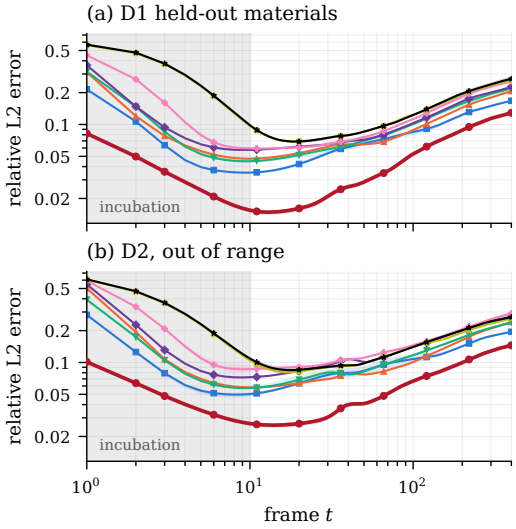
Figure 4: Duhamel-NO rolled out from unseen noise on two test materials, against FerroX. We show the top layer of the film. Each frame is scaled by the maximum of the true field, and the numbers are relative errors. Each material is the median-error trajectory of its group (D1 interior and D2 first-order wells). Mean errors are higher because of a tail of harder materials (Table 1). Figures 13 and 14 show every model.

7.2 The first frames

Figure 5a,b shows the error at every frame. The gain of Duhamel-NO is largest at the start. At frame 1 its error is 0.082, against 0.215 for F-FNO and 0.568 for FNO, and the gap persists to frame 400.

The truncation cap of Section 5 accounts for most of the FNO's early deficit (Figure 3). The trained FNO reaches a correlation of 0.826, 0.877 and 0.923 at frames 1–3, just below its cap of 0.853, 0.882 and 0.926. TFNO, which has the same mode set, behaves the same way. Duhamel-NO reaches 0.997 at frame 1. The cap is not the only source of error, however. At frame 1 every baseline's error is spread over all wavenumbers (Figure 11).

For every model, the error falls over the first ten frames and only then rises. This is unusual, because errors in an autoregressive rollout usually compound. For Duhamel-NO the reason is exact (Proposition 6 in Section A.5). Its Fourier SSM is the true linear map. An error on a decaying mode therefore shrinks, and an error on a growing mode grows only as fast as the field itself. For the baselines this is only a heuristic, since their own learned linear response does the damping. After nucleation, errors slowly accumulate for all models.



(c) Physical quantities, D1 held-out

	t_{nuc}	P_{S}	domain size	wall density	P/P_{S} hist.	φ	E_z	$F(t)$
Duhamel-NO	**0.28**	**0.003**	**0.005**	**0.007**	**0.010**	**0.139**	**0.135**	**0.004**
F-FNO	0.70	0.006	**0.005**	0.008	**0.010**	0.165	0.164	0.007
U-Net	0.61	0.008	**0.005**	0.008	0.011	0.201	0.200	0.006
UNO	0.54	0.024	0.006	0.009	0.020	0.220	0.218	0.012
CNextU-Net	0.55	0.005	0.006	0.008	**0.010**	0.224	0.222	0.007
CNO	0.46	0.006	0.008	0.010	0.011	0.256	0.253	0.008
TFNO	0.69	0.011	0.008	0.011	0.013	0.263	0.260	0.008
FNO	0.68	0.012	0.008	0.011	0.014	0.270	0.267	0.009

Figure 5: Error over time and physical quantities. (a,b) Relative error at every frame on (a) the D1 held-out materials and (b) D2. Lines are means over trajectories, bands show the range over seeds, and the shaded region is incubation. (c) Errors in physical quantities on the D1 held-out materials (mean over trajectories and seeds; one seed for U-Net, CNextU-Net and TFNO; **best**, second). t_{nuc} is in frames, $F(t)$ is relative to the total energy drop, and the others are relative errors at frame 400. φ is the electrostatic potential and E_z the depolarization field.

Table 2: Ablations of Duhamel-NO on the D1 held-out materials (mean ± std over seeds; columns as in Table 1). Rows (a), (b), (d) and (e) use the same 9.5M-parameter UNO. Rows (c) and (f) replace it with a small network whose output is exactly $O(v^2)$.

variant	D1 corr.	D1 error	frame-1 error
(a) UNO alone, no Duhamel step	0.962±0.003	0.219±0.012	0.315±0.005
(b) Duhamel-NO (measured Fourier SSM)	0.982±0.001	0.129±0.003	0.082±0.000
(c) Duhamel-NO, small $O(v^2)$ network instead of UNO	0.934±0.013	0.287±0.042	0.085±0.000
(d) Duhamel-NO, learned symbol instead of measured	0.950±0.001	0.256±0.003	0.399±0.026
(e) Duhamel-NO, uniform mode kept instead of damped	0.980	0.133	0.093
(f) learned symbol and small $O(v^2)$ network	0.812±0.002	0.578±0.003	0.219±0.001

7.3 Ablations

Table 2 isolates the parts of Duhamel-NO. Row (a) is the UNO alone, trained in the same way but without the Duhamel step. Adding the Duhamel step with the measured Fourier SSM, row (b), lowers the error from 0.219 to 0.129 at frame 400 and from 0.315 to 0.082 at frame 1. The other rows show which choices matter.

- *The Fourier SSM must be measured.* A learned, material-conditioned symbol (d) is worse than no Duhamel step at all, especially at frame 1. The learned symbol cannot couple the ten depth layers, while the measured 10×10 blocks do.
- *The correction must be flexible.* A small network whose output is exactly $O(v^2)$ (c) keeps frame 1 accurate, but it underfits the late coarsening.
- *Damping the uniform mode is a minor choice* (e).

7.4 Physical quantities

Figure 5c lists the physical quantities of Section 6.2, recomputed from each model's prediction on the D1 held-out materials. We ran paired tests over 12 metrics. These are the eight in Figure 5c, the field errors at frames 1, 10 and 400, and the wall energy (not shown). Duhamel-NO is significantly better than the best

baseline on 8 of them and tied on the other 4. The ties are late-time morphology statistics (domain size, wall density, wall energy and the P/P_S histogram). Duhamel-NO predicts the nucleation frame exactly for 73% of trajectories, against 62% for the best baseline.

7.5 Speed

On one dedicated A100 GPU, FerroX needs a median of 196 s per 400-frame D1 trajectory (range 147–318 s). Duhamel-NO needs 1.66 s for a single trajectory (118× faster) and 0.108 s per trajectory in batches of 16 (1814× the throughput). Simulating the 682 training trajectories takes about 37 GPU-hours, and training Duhamel-NO takes about 39 minutes. This cost is recovered after about 700 predicted trajectories. Figure 12 compares the cost and accuracy of all models.

7.6 Transfer to new grids

All FerroX data share one grid, so we test transfer to new grids on the 2D Allen–Cahn equation, which is the single-layer limit of TDGL without depolarization or a sextic term and is cheap to simulate on several grids (Section H). Trained on 64^2, Duhamel-NO runs unchanged on 128^2 with an error of 0.010, whereas F-FNO and FNO degrade to 0.086 and 0.276. This follows from its linear step, which depends on the dimensionless wavenumber $k\mu$ rather than on the grid index. On the coarser 32^2 grid, however, Duhamel-NO ranks only fourth (0.124, against 0.049 for F-FNO; Table 10).

8 Conclusion and Limitations

Duhamel-NO builds Duhamel's principle for the TDGL equation into a neural operator. The Fourier SSM, which decides the domain pattern, is computed in closed form from simulation data. Only the nonlinear correction is learned. As a result, Duhamel-NO is not limited by the truncation cap that bounds FNOs in the first frames. It generalizes zero-shot to unseen noise, held-out and out-of-range materials and first-order wells. It has lower error and better physical quantities than seven tuned baselines, at 1814× the throughput of the solver.

Limitations. Duhamel-NO is built for *noise-seeded dynamics*, where an unstable linear stage decides the pattern. On voltage-driven hysteresis, where the film spends most of its time near saturation, it is second to CNO (Section G). Its Fourier SSM must be measured from the first frames of the training simulations, so it needs simulations in which the linear stage is resolved. With only 32 training materials, F-FNO is more accurate (0.186 against 0.239; Figure 12). On Allen–Cahn, our data come from an exponential integrator, which favours our model, and Duhamel-NO is not the best model on a coarser grid. Finally, all ferroelectric data come from one simulator, with uniaxial polarization and one device stack, and none come from experiment.

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A Derivations and proofs

Throughout, $\hat{f}(k)$ is the in-plane Fourier transform of a field f, $k = (k_x, k_y)$, and a field on the $64 \times 64 \times 10$ grid is a vector of ten depth layers at every in-plane point. For a field u we write $\|u\|$ for the L_2 norm over all voxels.

A.1 The split of the TDGL equation

The free energy of Equation (1) is $F = F_2 + F_{46}$, with

$$F_2[P_z] = \int \Big[\tfrac{\alpha}{2}P_z^2 + \tfrac{G_{11}}{2}\|\nabla P_z\|^2\Big]d\mathbf{r} + U_{\text{elec}}[P_z], \qquad F_{46}[P_z] = \int \Big[\tfrac{\beta}{4}P_z^4 + \tfrac{\gamma}{6}P_z^6\Big]d\mathbf{r}.$$

The electrostatic energy is $U_{\text{elec}} = -\frac{1}{2}\int E_z[P_z]\,P_z\,d\mathbf{r}$, where $E_z[P_z]$ is the z-component of the field that solves the Poisson problem of the stack with P_z as its source. The Poisson problem is linear with fixed (grounded) boundary values, so $E_z[P_z]$ is linear in P_z and the map $P_z \mapsto -E_z[P_z]$ is symmetric. The variational derivatives are therefore

$$\frac{\delta F_2}{\delta P_z} = \alpha P_z - G_{11}\nabla^2 P_z - E_z[P_z], \qquad \frac{\delta F_{46}}{\delta P_z} = \beta P_z^3 + \gamma P_z^5,$$

and $\partial_t P_z = -\Gamma\,\delta F/\delta P_z$ is exactly $\partial_t P_z = \mathcal{L}P_z + \mathcal{N}(P_z)$ with $\mathcal{L}$ and $\mathcal{N}$ as in Equation (3). $\mathcal{L}$ is linear and self-adjoint (up to the factor $-\Gamma$), and it contains every coefficient that survives as $P_z \to 0$, namely α, G_{11} and the geometry of the stack. $\mathcal{N}$ contains only β and γ. This is why, in Duhamel-NO, the Fourier SSM depends on α alone, while β and γ enter only the correction (whose network also receives α through the material numbers).

A.2 The effective force, the exponential-Euler step and the filter φ_1

Section 4.3 shows that $\tau\varphi_1(\tau\mathcal{L})$ is symmetric positive definite, so the effective force of Equation (5) is well defined, and that it equals $\int_0^\tau e^{(\tau-s)\mathcal{L}}ds$. Below we also use that $\varphi_1(z) = \int_0^1 e^{zr}dr$ is increasing on the real line, and that, by Equation (4), $\tau\varphi_1(\tau\mathcal{L})N_{\text{eff}}(P_z(t)) = \int_0^\tau e^{(\tau-s)\mathcal{L}}\mathcal{N}(P_z(t+s))\,ds$.

Distance from the frozen force. Freezing the integrand of Equation (4) at $s = 0$ gives

$$P_z(t+\tau) \approx e^{\tau\mathcal{L}}P_z(t) + \tau\,\varphi_1(\tau\mathcal{L})\,\mathcal{N}(P_z(t)),$$

the exponential-Euler scheme (Cox & Matthews, 2002; Hochbruck & Ostermann, 2010), so $N_{\text{eff}} = \mathcal{N}(P_z(t))$ if $\mathcal{N}$ is constant over the frame. In general,

$$\tau\varphi_1(\tau\mathcal{L})\big[N_{\text{eff}}(P_z(t)) - \mathcal{N}(P_z(t))\big] = \int_0^\tau e^{(\tau-s)\mathcal{L}}\big[\mathcal{N}(P_z(t+s)) - \mathcal{N}(P_z(t))\big]ds,$$

and the norm of the right-hand side is at most $\frac{\tau^2}{2}\sup_{s\le\tau}\|e^{(\tau-s)\mathcal{L}}\|\,\sup_{s\le\tau}\|\frac{d}{ds}\mathcal{N}(P_z(t+s))\|$. On a mode with $\tau\mu \ge -1$, $\tau\varphi_1(\tau\mu) \ge \tau\varphi_1(-1) = 0.63\,\tau$, so on such modes $N_{\text{eff}} - \mathcal{N}(P_z(t)) = O(\tau)$. It vanishes when $\mathcal{N}$ does not change within a frame, and in particular during incubation, where $\mathcal{N} = O(P_z^3)$. On stiff modes ($\tau\mu \ll -1$), $\tau\varphi_1(\tau\mu) \approx 1/|\mu|$, so any component of N_{eff} on those modes enters the step with a small weight, and the network does not need to resolve it. Duhamel-NO replaces N_{eff} by a learned N_ψ, trained so that the whole step matches the data. It learns the effective force along the path, not the frozen value at $s = 0$.

A.3 Proof that the linear dynamics are a diagonal Fourier SSM

Proof of Proposition 3. Let $\Lambda(k)$ be diagonal and set $h^t(k) = V(k)^{-1}\hat{P_z}^{\,t}(k)$. Then $h^{t+1} = V^{-1}B\hat{P_z}^{\,t} = V^{-1}V\Lambda V^{-1}\hat{P_z}^{\,t} = \Lambda h^t$ and $\hat{P_z}^{\,t} = Vh^t$. This is the state-space recurrence $h^{t+1} = Ah^t + Bx^t$, $y^t = Ch^t + Dx^t$ with $A = \Lambda$, $C = V$, $D = 0$ and an input x^t that is the initial noise at $t = 0$ and zero afterwards. Every

mode k is an independent copy of this system, so the full linear dynamics are a bank of diagonal, linear, time-invariant SSMs, one per in-plane mode. □

Remark. Diagonalizability holds generically and is verified numerically. On the reference material the relative reconstruction error $\|V\Lambda V^{-1} - B\|_\infty / \|B\|_\infty$ is 2.9×10^{-15}, with condition number $\kappa(V) \leq 6.39$. In the continuum the generator is self-adjoint up to the mobility (Section A.1), so its eigenvalues are real; in the fitted discrete map all 128 unstable eigenvalues of the reference material are real to 3.9×10^{-8}, and complex pairs occur only among modes that decay within two frames.

A.4 Material units and the shared nonlinearity

Write $P_z = P_S v$ and measure time in units of λ. Divide $\mathcal{N}$ by $\Gamma|a_{\text{sat}}|P_S$, where $|a_{\text{sat}}| = \beta P_S^2 + \gamma P_S^4$ defines P_S (Section C.3):

$$\frac{\mathcal{N}(P_S v)}{\Gamma|a_{\text{sat}}|P_S} = -\frac{\beta P_S^2}{|a_{\text{sat}}|} v^3 - \frac{\gamma P_S^4}{|a_{\text{sat}}|} v^5 = -(1-r)\, v^3 - r\, v^5, \qquad r = \frac{\gamma P_S^4}{|a_{\text{sat}}|}.$$

The second equality uses $\beta P_S^2 / |a_{\text{sat}}| = 1 - r$, which follows from the definition of P_S. So in material units every material has the same nonlinearity up to the single number r, and the corresponding reduced potential $\phi_r(v) = -v^2/2 + (1-r)v^4/4 + rv^6/6$ has its minima at $v = \pm 1$ for every r (Figure 7). $r \in [0, 1)$ is a second-order well, $r = 1$ means $\beta = 0$, and $r > 1$ is a first-order well ($\beta < 0$), held only by the sextic term.

The shape is shared, but the size is not quite. Time is measured in $\lambda = 1/(\Gamma|a_{\text{rate}}|)$, not in $1/(\Gamma|a_{\text{sat}}|)$, so in material units the force is $(|a_{\text{sat}}|/|a_{\text{rate}}|)\left(-(1-r)v^3 - rv^5\right)$. The two coefficients have different critical points (Section C.3), so the prefactor $|a_{\text{sat}}|/|a_{\text{rate}}|$ varies across materials. It is a function of α and r, which the network receives through the material numbers $\mathbf{m}$ (α through Π_h), so the network can absorb it.

A.5 Why the error falls during incubation

The following statement makes the argument of Section 7.2 precise for the linear stage. Let the true field obey $P_{n+1} = BP_n$ with B diagonalizable, $B = V\Lambda V^{-1}$, and split the eigen-coordinates into an unstable set $\mathcal{U}$ ($|\lambda_i| > 1$) and a stable set $\mathcal{S}$ ($|\lambda_i| \leq \rho_s < 1$). Let a surrogate produce $\tilde{P}_{n+1} = B\tilde{P}_n + \delta_n$, where δ_n is the error it adds at step n.

Proposition 6 (error propagation in the linear stage)**.** The error $e_n = \tilde{P}_n - P_n$ satisfies $e_n = B^n e_0 + \sum_{j<n} B^{n-1-j}\delta_j$. In eigen-coordinates $c = V^{-1}e$, the stable components of the carried-over error decay geometrically, $|c_i(B^n e_0)| \leq \rho_s^n |c_i(e_0)|$ for $i \in \mathcal{S}$, while the unstable components grow at exactly the rate of the corresponding components of the true field. Consequently, if the injected errors are small after the first step ($\|\delta_j\| \ll \|e_0\|$ for $j \geq 1$), the relative error $\|e_n\|/\|P_n\|$ decreases with n as long as the stable part of e_0 dominates the error and the field is dominated by its unstable part.

Proof. The recursion follows by induction. In eigen-coordinates B^n acts as Λ^n, so component i of $B^n e_0$ is $\lambda_i^n c_i(e_0)$. It shrinks as ρ_s^n on $\mathcal{S}$ and grows as $|\lambda_i|^n$ on $\mathcal{U}$, the same factor that multiplies component i of $P_n = B^n P_0$. The ratio of unstable error to unstable field is therefore constant, while the stable error decays and the stable part of the field decays too, but the field norm is increasingly carried by $\mathcal{U}$. □

For Duhamel-NO in the linear stage, the hypotheses of Proposition 6 hold with the true rates, because the Fourier SSM of Equation (7) is the measured B, and the correction is $O(\|v\|)$ times a small learned term. For the baselines the proposition does not apply directly, because their linear response is learned and is not exactly B. The same argument holds for them only to the extent that their own learned response damps the decaying modes.

A.6 The truncation cap: definitions and proofs

This section proves Propositions 4 and 5 of Section 5. It also adds two results on how the pattern correlation behaves as the noise shrinks. Write the one-frame map as $S(u) = Au + R(u)$, with A a Fourier multiplier $g(k)$ (a $Z \times Z$ matrix per wavenumber) and $\|R(u)\| \leq C\|u\|^p$, $p \geq 2$. A field F with its spatial mean removed is written $\tilde{F}$, and $\rho(F,T) = \mathbb{E}\langle \tilde{F}, \tilde{T}\rangle / \sqrt{\mathbb{E}\|\tilde{F}\|^2\, \mathbb{E}\|\tilde{T}\|^2}$ is the population pattern correlation. Let $J_1 F$ be the

orthogonal projection of the random field F onto the linear span of the starting-noise components. This is the part of F that is linear in the noise (for Gaussian noise, the first Wiener chaos (Nualart, 2006); Gaussianity is not needed). Let $\mathcal{H}(F) = 1 - \mathbb{E}\|J_1\tilde{F}\|^2/\mathbb{E}\|\tilde{F}\|^2$ be the share of the output that is not linear in the noise, and $\rho_{\text{lin}}(F,T) = \rho(J_1F,T)$.

Proposition 7 (exact factorization)**.** Let $T = A\epsilon\xi$ be the true first frame from noise $\epsilon\xi$. For every surrogate output $G = G(\epsilon\xi)$ with finite second moment, $\rho(G,T) = \rho_{\text{lin}}(G,T)\sqrt{1-\mathcal{H}(G)} \le \sqrt{1-\mathcal{H}(G)}$.

Proof. $\tilde{T}$ lies in the linear span of the noise components and $\tilde{G} - J_1\tilde{G}$ is orthogonal to that span, so $\mathbb{E}\langle\tilde{G},\tilde{T}\rangle = \mathbb{E}\langle J_1\tilde{G},\tilde{T}\rangle$ and $\mathbb{E}\|\tilde{G}\|^2 = \mathbb{E}\|J_1\tilde{G}\|^2 + \mathbb{E}\|\tilde{G} - J_1\tilde{G}\|^2$. Hence $\rho = \rho_{\text{lin}}\sqrt{\mathbb{E}\|J_1\tilde{G}\|^2/\mathbb{E}\|\tilde{G}\|^2} = \rho_{\text{lin}}\sqrt{1-\mathcal{H}}$, and $|\rho_{\text{lin}}| \le 1$ by Cauchy–Schwarz. □

Proposition 8 (three behaviours as the noise shrinks)**.** (a) If G is Lipschitz with a non-zero, spatially varying output at zero input (biases with positional channels), then $\mathcal{H}(G(\epsilon\xi)) \to 1$ and $\rho \to 0$ as $\epsilon \to 0$. (b) If $G(u) = r(u)\Phi(u/r(u), \log r(u))$ with $r = \text{rms}(u)$, then $\mathcal{H}$ depends on ϵ only through the explicit $\log r$ input. (c) If $G(u) = \hat{A}u + N(u)$ with $\|N(u)\| \le K\|u\|^2$, then $\mathcal{H} = O(\epsilon^2)$ and $\rho \to \rho_{\text{lin}}(\hat{A}, A)$.

Proof. (a) Let L be the Lipschitz constant, $b = \tilde{G}(0) \ne 0$ and $c = L\epsilon(\mathbb{E}\|\xi\|^2)^{1/2}$. Then $\mathbb{E}\|\tilde{G}(\epsilon\xi) - b\|^2 \le c^2$, so the mean part satisfies $\|J_0\tilde{G}\| \ge \|b\| - c$ and $\mathbb{E}\|\tilde{G}\|^2 \le (\|b\| + c)^2$, giving $\mathcal{H} \ge (\|b\| - c)^2/(\|b\| + c)^2 \to 1$; and $\mathbb{E}\|J_1\tilde{G}\|^2 \le c^2$ bounds $\rho \le c/(\|b\| - c) \to 0$. (b) $\mathcal{H}$ is invariant under deterministic rescaling, so $\mathcal{H}(G(\epsilon\xi)) = \mathcal{H}(r(\xi)\Phi(\xi/r(\xi), \log\epsilon + \log r(\xi)))$. (c) $\hat{A}\epsilon\xi$ is linear in the noise, so the non-linear share comes only from N, with $\mathbb{E}\|N\|^2 \le K^2\epsilon^4\mathbb{E}\|\xi\|^4$, while $\mathbb{E}\|\tilde{G}\|^2 \ge (\epsilon\|\hat{A}\tilde{\xi}\| - K\epsilon^2(\mathbb{E}\|\xi\|^4)^{1/2})^2$. Hence $\mathcal{H} = O(\epsilon^2)$, and the linear part of N is $O(\epsilon^2)$, so $\rho_{\text{lin}}(G) \to \rho_{\text{lin}}(\hat{A})$. □

Proof of Proposition 4. At a homogeneous state every pre-activation is spatially constant, so layer l has Jacobian $D_l(W_l + R_l(k)\mathbb{1}[k \in \mathcal{K}])$ with $D_l = \text{diag}(\phi'(\cdot))$. Products of multipliers are multipliers, and outside $\mathcal{K}$ only the k-independent W_l terms remain. With coordinate channels the action above the cut-off is multiplication by a fixed smooth field, so the statement holds up to the low-wavenumber content of that field. □

Proof of Proposition 5. We show the slightly stronger statement $\rho_{\text{lin}}(G, A^n) \le \rho_{\mathcal{K}}(n)$, which gives $\rho(G, T_n) \le \rho_{\mathcal{K}}(n)\sqrt{1-\mathcal{H}(G)} \le \rho_{\mathcal{K}}(n)$ by Proposition 7. For translation-equivariant G and white noise, $J_1G = \mathcal{F}^{-1}[m(k)\mathcal{F}\xi]$, because linear regression on white noise is a convolution. So ρ_{lin} is the spectral cosine between m and g_n. Maximizing it over m with $m \equiv c$ on $\mathcal{K}^c$ gives $m = g_n$ on $\mathcal{K}$ and c equal to the mean of g_n on $\mathcal{K}^c$. Substituting gives $\rho_{\mathcal{K}}(n)$. □

By Proposition 4, the linear response of an FNO is constant outside its mode set, so Proposition 5 applies to it. For rms-normalized networks (class (b) of Proposition 8), such as our trained baselines, the cap is an empirical reference, not a proven bound. In its linear part, Duhamel-NO is of class (c). Its Fourier SSM acts on every mode, so no cap applies.

B Algorithms

Algorithm 1 (Section 4) rolls out Duhamel-NO from the initial noise. The three procedures below measure the Fourier SSM (Algorithm 2), build it for a new material (Algorithm 3), and train the network (Algorithm 4).

Algorithm 2 Measuring the Fourier SSM of one training material

Require: frames $P^0, \dots, P^8$ of one training simulation (field still small); frame interval θ (in growth times)

$\hat{P}^t \leftarrow$ in-plane FFT of P^t for $t = 0, \dots, 8$ ▷ $\hat{P}^t(k) \in \mathbb{C}^{10}$, one vector per mode

for each radial bin b of $|k|$ (46 bins) **do**

  $X \leftarrow [\hat{P}^t(k)]_{k\in b,\ t<8}$, $Y \leftarrow [\hat{P}^{t+1}(k)]_{k\in b,\ t<8}$ ▷ snapshot pairs

  $B_b \leftarrow YX^{\mathsf{H}}(XX^{\mathsf{H}} + \epsilon I)^{-1}$ ▷ least squares (small ridge), 10×10

  $L_b \leftarrow \theta^{-1}\log B_b$ ▷ generator, symmetrized

bins whose modes decay below round-off within one frame get a learnable, strongly damped rate; the uniform mode is damped

return $\{L_b\}$ and $\log|\alpha|$ of this material

Algorithm 3 The Fourier SSM of a new material

Require: α of the new material; generators $\{L_b^{(i)}\}$ of all training materials i

$\mathcal{I} \leftarrow$ the eight training materials with $\log|\alpha_i|$ closest to $\log|\alpha|$

$L_b \leftarrow \frac{1}{8}\sum_{i\in\mathcal{I}} L_b^{(i)}$ for every bin b ▷ β, γ are not needed, since they do not enter $\mathcal{L}$

return $L(k) = L_{b(|k|)}$, and $e^{\theta L(k)}$, $\theta\varphi_1(\theta L(k))$ by `matrix_exp`

Algorithm 4 Training Duhamel-NO

Require: training trajectories (682 materials, one trajectory each); Fourier SSMs from Algorithm 2

initialize N_ψ with a zero read-out ▷ the untrained model is the exact linear solution

for 20 000 steps **do**

  sample 16 windows of $k+1 = 5$ frames; with probability 0.6 the window starts in frames 0–120 (growth phase)

  apply a random exact symmetry (90° rotation, reflection or $P_z \to -P_z$)

  roll out $k = 4$ steps with Algorithm 1 from the first frame of the window (gradients flow through all steps)

  $\mathcal{L} \leftarrow \frac{1}{k}\sum_{q=1}^{k} \|\tilde{P}^q - P^q\|^2 / \|P^q\|^2$ ▷ relative L_2

  AdamW (Adam with decoupled weight decay) step (one-cycle schedule, weight decay 10^{-4}, gradient clipping at 1)

C Datasets and material units

C.1 Simulations

All data are FerroX (Kumar et al., 2023) simulations of the stack of Section 6.1, with explicit-Euler TDGL steps and a Poisson solve for the whole stack at every step. Each run starts from a field drawn uniformly from $\pm 2 \times 10^{-3}\,\mathrm{C/m^2}$ in every cell, which is 0.5–2% of P_S for the D1 materials and 0.2–4% for D2. The field is set by an integer seed (D1 material i uses seed $2000 + i$, and D2 uses $6000 + i$). A trajectory is rejected if it has non-finite values or $|P_z| > 1.35 P_\mathrm{S}$; none was rejected.

Table 3: Datasets. Every trajectory has its own noise seed; no seed is shared between training and test data. Hysteresis is described in Section G.

dataset	grid	frames	varied parameters	training	test
D1	$64^2 \times 10$	401	α, β, γ (Latin hypercube (McKay et al., 1979) in $\log\|\alpha\|, P_\mathrm{S}, r$)	682 (+63 validation)	256 held-out materials (82 interior, 174 edge)
D2	$64^2 \times 10$	401	α, β, γ drawn independently, five regions	–	300 (5×60), no retraining
Hysteresis	$64^2 \times 10$	440–2765	α, β, γ, peak voltage	90 loops (+10 validation)	30 held-out material, 34 new noise (12 evaluated), 10 closed set

Table 3 summarizes the datasets, and Figure 6 shows where the D1 and D2 materials lie in $(|\alpha|, P_\mathrm{S}, r)$.

D1. The coefficients are $\gamma = r|a_\mathrm{sat}|/P_\mathrm{S}^4$ and $\beta = (1-r)|a_\mathrm{sat}|/P_\mathrm{S}^2$, so every sample is a valid double well. Sampling the raw coefficients instead would reject more than 95% of draws, because the admissible range of β moves by two orders of magnitude across the range of α. The validation materials are every twelfth training material.

D2. 300 materials whose α, β, γ are drawn independently (log-uniform) and assigned to five regions of 60. **A** has small P_S. $\mathbf{B}_\mathrm{units}$ has large P_S, up to 1.8× the training maximum (a pure change of units, because the dimensionless numbers stay in range). $\mathbf{B}_\alpha$ has large $|\alpha|$ with small P_S. **C** has first-order wells with $r = 1.15$–2 (a change of a dimensionless number, i.e. new physics). **D** combines all of these, with P_S up to 2.2× the training maximum.

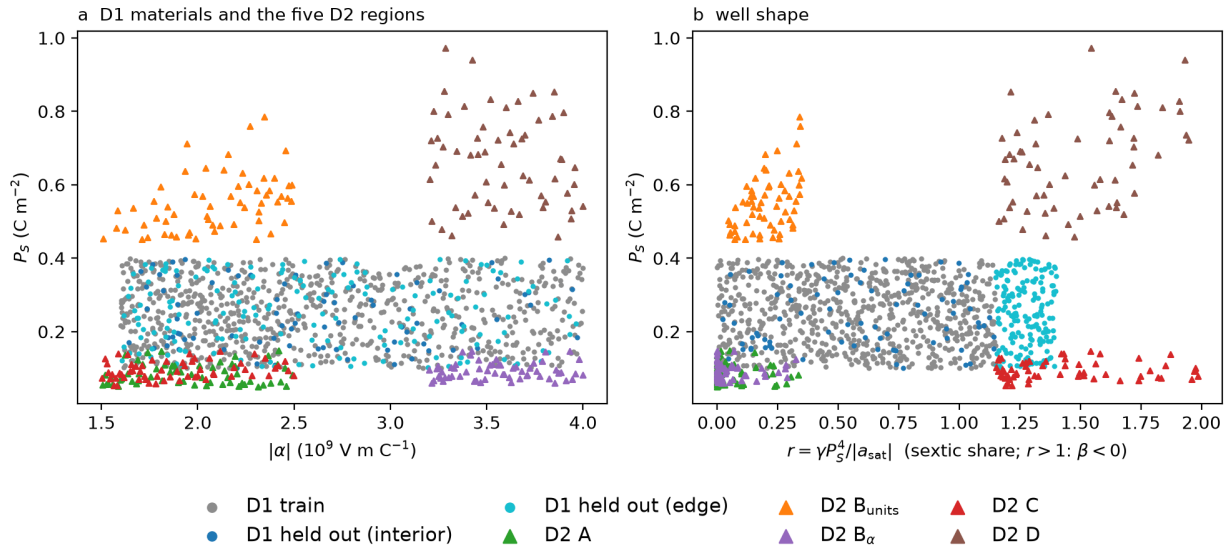


Figure 6: Where the materials lie. D1 materials (circles, grey for training and blue for held out) and D2 materials (triangles, coloured by region) in well depth $|\alpha|$, saturation polarization P_S and well shape r. D2 leaves the training range in P_S and in r.

C.2 Free-energy landscapes

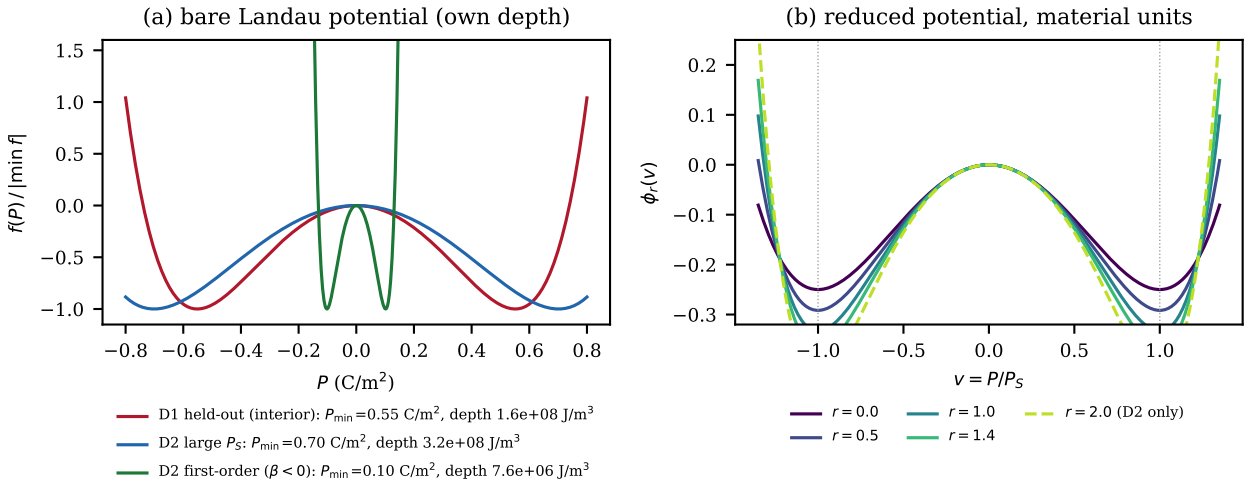


Figure 7: Free-energy landscapes of the test materials. (a) Bare Landau potential $f(P) = \alpha P^2/2 + \beta P^4/4 + \gamma P^6/6$ of the two materials of Figure 4 and the D2 large-P_S material, each divided by its own well depth. The wells differ in position by a factor of seven and in depth by a factor of about forty, and the first-order material ($\beta < 0$) has a narrow, steep well. (b) In material units every material has the reduced potential $\phi_r(v)$ of Section A.4, with minima at $v = \pm 1$; only its shape, set by r, differs. D2 reaches $r = 2$, beyond the training range (dashed).

C.3 Material units

The depolarization field of the dielectric renormalizes α, and it does so differently for the two things α controls, namely the growth of small fluctuations and the saturated amplitude. Fitting simulations gives a growth coefficient $|a_{rate}| \propto |\alpha| - 1.37 \times 10^9$, which sets the growth time λ, and a saturation coefficient $|a_{sat}| = 1.389\,(|\alpha| - 1.31 \times 10^9\,(1 - 0.0775\,r))$ (units of α, V m/C). The saturated polarization P_S solves

$|a_{\mathrm{sat}}| = \beta P_{\mathrm{S}}^2 + \gamma P_{\mathrm{S}}^4$, and the shape number is $r = \gamma P_{\mathrm{S}}^4 / |a_{\mathrm{sat}}|$. This law predicts P_{S} of the D1 materials to 1.03% on average (3.41% at worst) and of the D2 materials, whose P_{S} reaches 2.2× the fit maximum, to 1.9% (Figure 8). The value 1.37×10^9 is a critical point. For $|\alpha|$ below it the depolarization field outweighs the Landau driving force and the film never nucleates; all D1 and D2 materials lie above it. Over 120 D1 materials, the change of units also reduces the coefficient of variation of the settled amplitude from 0.38 to 0.083.

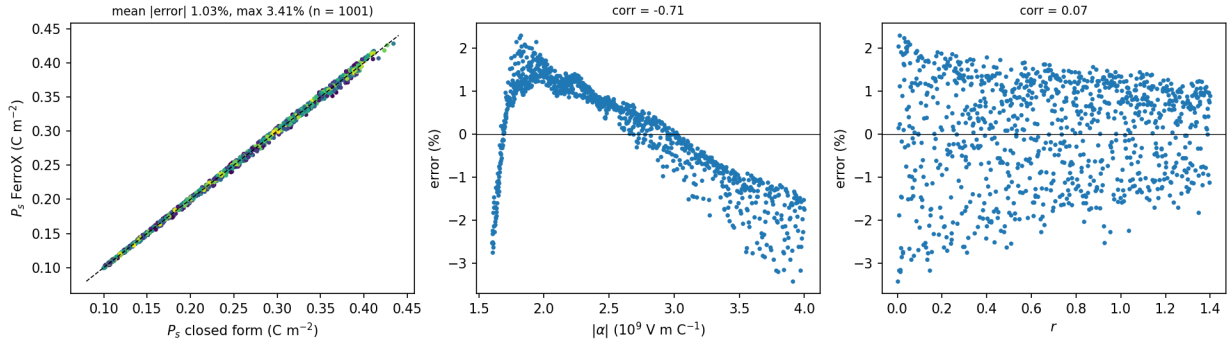


Figure 8: Closed-form saturation polarization against the value measured in FerroX (99.5% quantile of $|P|$ over the settled frames) for all D1 materials, coloured by r, with residuals against $|\alpha|$ and r.

D Training and evaluation details

Architecture of N_ψ. Each UNO block multiplies the lowest Fourier modes of its input by learned weights, adds a pointwise linear map, applies a Gaussian error linear unit (GELU), and resamples the field to a coarser or finer grid. The input has four parts: the field divided by its rms amplitude s, a channel holding $\log_{10} s$, a 16-dimensional embedding of $\mathbf{m}$ (a bias-free two-layer network), and two coordinate channels. A pointwise layer lifts these to 32 channels. Two down-sampling blocks reduce the grid from 64^2 to 32^2 and 16^2 (64 and 128 channels), a middle block works at 16^2, and two up-sampling blocks return to 64^2 with skip connections. A final block and a two-layer pointwise head give ten output channels, which are multiplied back by s. The read-out starts at zero.

Training. Every model uses 20 000 optimizer steps, batch 16, $k = 4$ unrolled steps per loss evaluation, AdamW (Loshchilov & Hutter, 2019) (weight decay 10^{-4}) with a one-cycle schedule (Smith & Topin, 2019) (5% warm-up), gradient clipping at 1, 32-bit floating point (fp32), and exact symmetry augmentation. Besides the seven baselines, we trained a transformer, Transolver (Wu et al., 2024), and three generative models on a U-Net backbone. The generative models are trained on their own objectives (one-step denoising, or flow matching on the residual). They are sampled with 20 denoising diffusion implicit model (DDIM) (Song et al., 2021) steps for the autoregressive conditional diffusion model (ACDM) (Kohl et al., 2023), 8 Euler steps for flow matching (Lipman et al., 2023), and 3 refinement steps for PDE-Refiner (Lippe et al., 2023). Poseidon-T is fine-tuned from its public pretrained weights for 20 000 steps.

Without our training setup. Every baseline is trained with the setup of Sections 4.5 and 4.6. Without it, a baseline receives the raw material parameters (standardized) as constant input channels and periodic coordinate channels. Table 4 shows that every backbone needs the setup.

Table 4: Standard training against our training setup, same backbone, same (4000-step) tuning budget, FerroX validation split (new materials, new noise). The best standard configuration is a relative-L_2 loss with per-sample normalization for every backbone except Transolver* (relative-L_2 loss alone).

	best standard configuration		with our training setup	
backbone	corr@400	rel. L2@400	corr@400	rel. L2@400
F-FNO	0.837	2.92	0.895	0.41
U-Net	0.841	2.04	0.909	0.37
UNO	0.842	1.87	0.863	0.47
CNextU-Net	0.742	1.52	0.853	0.49
CNO	0.622	1.00	0.907	0.38
TFNO	0.678	2.18	0.863	0.50
FNO	0.819	1.79	0.849	0.52
Transolver*	0.044	1.76	0.066	1.30

Selection. Learning rate and configuration are chosen per model on the validation split (Table 5 lists the result for every model and task), by correlation minus 0.1 times the relative error (for hysteresis, the loop error). Variants of U-Net, CNextU-Net and CNO without coordinate channels (and CNO with batch normalization, BatchNorm) are admitted as candidates. The chosen configuration is retrained with three seeds (one for the slowest models).

Evaluation choices. Predictions are clamped to three times the largest training value of the field, scaled per sample by its amplitude unit, so that out-of-range materials are not cut off. The neuralop FNO with `n_modes=(16,16)` keeps modes up to 8 in each direction; the truncation cap (Proposition 5) uses this true mode set. Timing uses one A100 with nothing else running.

Reproducibility. Every number in the paper and in this appendix is written as a LaTeX macro or a generated table from the benchmark result files (`make_duhamel_macros.py`, `make_physics_table.py`, `appendix/copy_assets.py`); none is typed by hand. The rollout figures are produced from the evaluated checkpoints by `make_rollout_fig.py` and `appendix/gallery_plot.py`.

Table 5: Validation-selected configuration, size, number of seeds and training time (one A100) of every model on every task. MSE is the mean squared error loss.

task	model	params (M)	seeds	train (min)	lr	configuration
FerroX TDGL	Duhamel-NO (ours)	9.49	3	39	0.0003	our setup
	F-FNO	1.19	3	42	0.001	our setup
	U-Net	10.95	1	76	0.0003	our setup
	UNO	9.49	3	27	0.001	our setup
	CNextU-Net	9.93	3	59	0.0003	our setup
	CNO	5.50	3	93	0.001	our setup
	TFNO	1.03	3	27	0.001	our setup
	FNO	2.42	3	27	0.001	our setup
	Transolver	2.19	1	161	0.001	our setup
	Poseidon-T	20.79	1	129	0.0001	rel. L2
	U-Net + ACDM diffusion	10.94	1	16	0.001	global noise scale
	U-Net + flow matching	10.94	1	23	0.001	global noise scale
	U-Net + PDE-Refiner	10.94	1	25	0.001	global noise scale
hysteresis	Duhamel-NO (ours)	9.49	3	81	0.001	our setup
	F-FNO	1.19	1	53	0.001	our setup
	U-Net	10.94	1	52	0.0003	rel. L2
	UNO	9.49	1	22	0.001	our setup
	CNextU-Net	9.93	3	43	0.0003	our setup
	CNO	5.50	3	74	0.0003	our setup
	TFNO	1.03	1	24	0.001	MSE
	FNO	2.42	1	24	0.001	our setup

E Additional results on ferroelectric domain formation

Table 6 gives every model with standard deviations over seeds, and Table 7 and Figure 9 break the results down by test subset. Duhamel-NO is the most accurate model on every subset, inside and outside the training range.

Table 6: All models (mean ± std over seeds; **best**). Pattern correlation and relative L_2 error at frame 400 on D1 (held-out materials) and D2 (out of range), and relative error at frame 1 (D1).

			D1 held-out (256)		D2 out-of-range (300)		frame 1
model	family	seeds	corr@400	rel. L2@400	corr@400	rel. L2@400	rel. L2
Duhamel-NO (ours)	ours	3	**0.982**±0.001	**0.129**±0.003	**0.978**±0.001	**0.145**±0.004	**0.082**±0.000
F-FNO	spectral NO	3	0.974±0.001	0.167±0.003	0.968±0.000	0.195±0.004	0.215±0.001
U-Net	U-Net	1	0.966	0.207	0.954	0.242	0.313
UNO	spectral NO	3	0.962±0.003	0.219±0.012	0.959±0.002	0.238±0.008	0.315±0.005
CNextU-Net	U-Net	3	0.959±0.000	0.229±0.002	0.949±0.002	0.261±0.010	0.347±0.018
CNO	conv. NO	3	0.951±0.004	0.256±0.011	0.937±0.004	0.292±0.005	0.451±0.003
TFNO	spectral NO	3	0.949±0.005	0.272±0.017	0.952±0.003	0.265±0.012	0.564±0.004
FNO	spectral NO	3	0.947±0.004	0.270±0.015	0.947±0.003	0.268±0.012	0.568±0.003
Poseidon-T	foundation	1	0.553	2.917	0.414	2.570	4.342
U-Net + ACDM diffusion	diffusion	1	0.206	1.293	0.092	2.854	5.209
U-Net + flow matching	flow matching	1	0.169	1.330	0.094	1.895	5.210
U-Net + PDE-Refiner	diffusion	1	0.053	3.578	0.033	2.158	5.351
Transolver	transformer	1	0.003	1.329	0.008	1.313	0.757

Table 7: Correlation at frame 400 per test subset (mean ± std over seeds). D1 is split into interpolation (inside the sampled box) and extrapolation (at its edges). The D2 regions are those of Section C.

model	D1 extrap.	D1 interp.	D2-A	D2-B_α	D2-B_{units}	D2-C	D2-D
Duhamel-NO (ours)	0.981±0.001	0.982±0.001	0.974±0.001	0.977±0.002	0.985±0.002	0.974±0.001	0.978±0.003
F-FNO	0.975±0.001	0.973±0.001	0.967±0.002	0.958±0.002	0.982±0.001	0.970±0.003	0.961±0.001
U-Net	0.963	0.972	0.967	0.970	0.979	0.941	0.915
UNO	0.959±0.004	0.970±0.002	0.958±0.010	0.949±0.003	0.976±0.004	0.951±0.008	0.960±0.003
CNextU-Net	0.958±0.000	0.962±0.001	0.951±0.008	0.957±0.004	0.969±0.002	0.941±0.003	0.926±0.016
CNO	0.949±0.004	0.955±0.007	0.917±0.008	0.952±0.011	0.959±0.004	0.923±0.018	0.934±0.009
TFNO	0.946±0.005	0.954±0.006	0.958±0.004	0.942±0.007	0.973±0.006	0.954±0.004	0.932±0.007
FNO	0.944±0.006	0.954±0.003	0.952±0.002	0.927±0.004	0.965±0.002	0.949±0.005	0.941±0.014

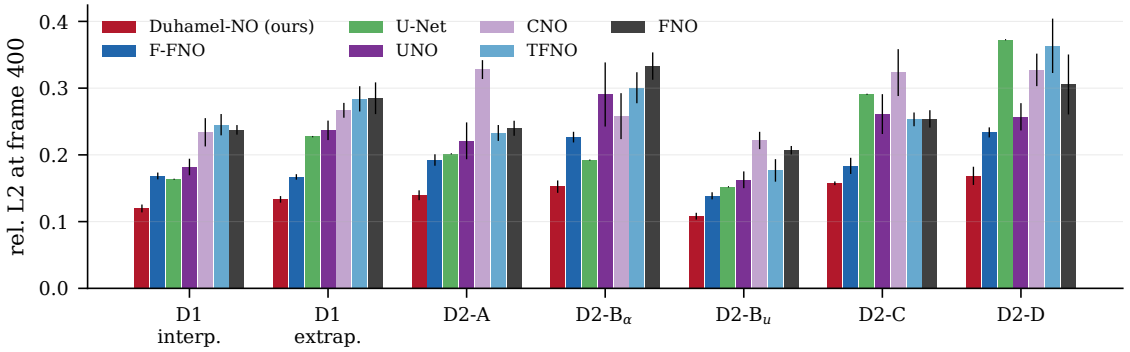


Figure 9: Relative error at frame 400 per test subset (bars show the standard deviation over seeds). Duhamel-NO is the most accurate model on every subset, including those outside the training range.

E.1 Physical quantities

Figure 10 shows the physical metrics of Figure 5c on D1 in more detail.

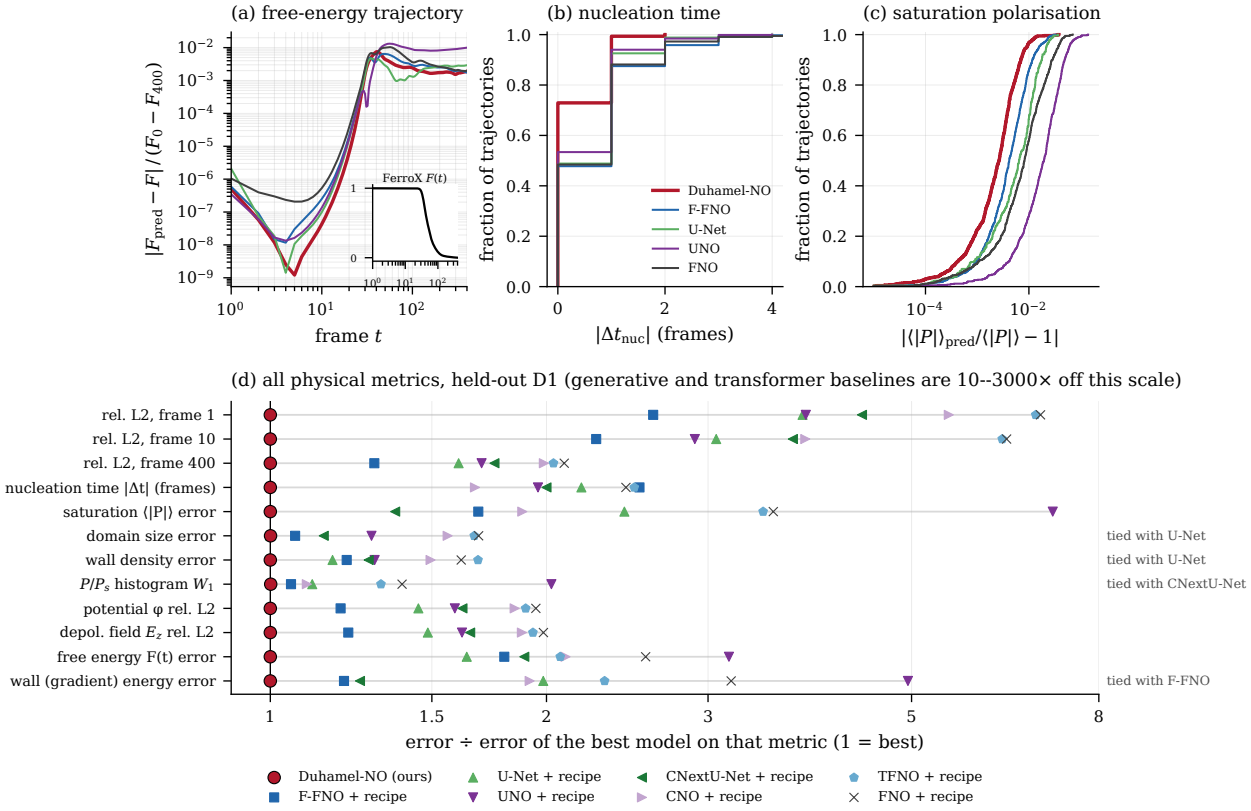


Figure 10: Physical consistency on the D1 held-out materials. (a) Error of the free-energy trajectory relative to the total energy drop (the inset shows the true $F(t)$). (b, c) Cumulative distributions of the nucleation-time error and of the saturation-polarization error (further left is better). (d) Every metric as the ratio of each model's error to that of the best model. In legends, "+ recipe" denotes our training setup.

E.2 Error spectra

Figure 11 shows where in wavenumber each model's error lies. At frame 1, where the truncation cap (Proposition 5) binds, every baseline's error is spread over all wavenumbers, so the cap is not the only source of error. By frame 10 the FNO's error is concentrated above its mode cut-off, as its constant response there predicts, but by then the cap itself is no longer restrictive (0.996).

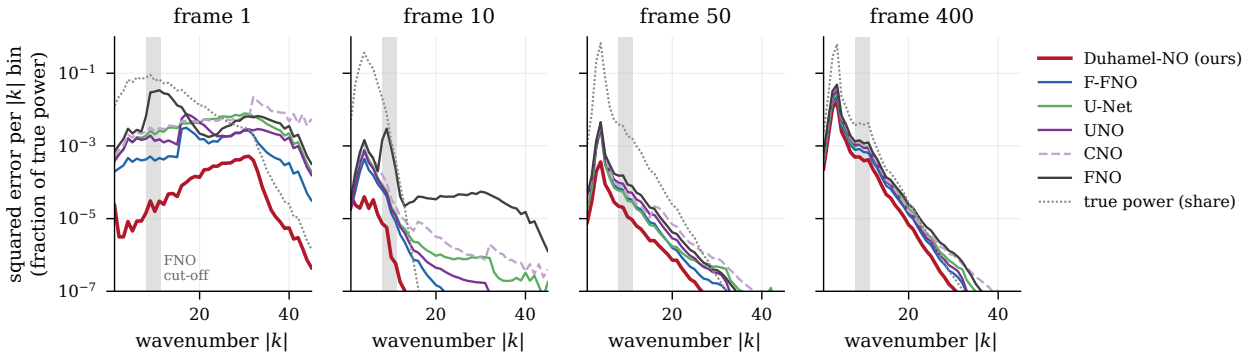


Figure 11: Error spectra on D1 held-out materials, as the squared error per radial wavenumber bin, as a fraction of the total true power, at four frames. The grey dotted line is the true power spectrum, and the shaded region is the FNO's mode cut-off. At frame 1 every baseline's error is spread over all wavenumbers; by frame 10 the FNO's error is concentrated above its cut-off.

E.3 The change of units

Table 8 isolates the change of units. The ablations of the Duhamel step are in Table 2.

Table 8: The change of units (mean ± std over three seeds) for FNO and UNO with our training setup, with and without material units. B_{units} are D2 materials with P_S up to 1.8× the training maximum. The units carry the amplitude out of range (0.24 → 0.02 for FNO, 0.26 → 0.02 for UNO); they do not improve the pattern in range.

backbone	change of units	D1 rel. L_2	D1 amplitude err.	D2 B_{units} rel. L_2	D2 B_{units} amplitude err.
FNO	yes	0.270 ±0.015	0.012 ±0.004	0.207 ±0.006	0.015 ±0.005
FNO	no	0.245 ±0.008	0.018 ±0.000	0.335 ±0.009	0.243 ±0.007
UNO	yes	0.219 ±0.012	0.025 ±0.007	0.163 ±0.013	0.019 ±0.002
UNO	no	0.199 ±0.004	0.049 ±0.002	0.330 ±0.006	0.258 ±0.002

E.4 Cost and training data

Figure 12 relates accuracy to inference cost and to the number of training materials.

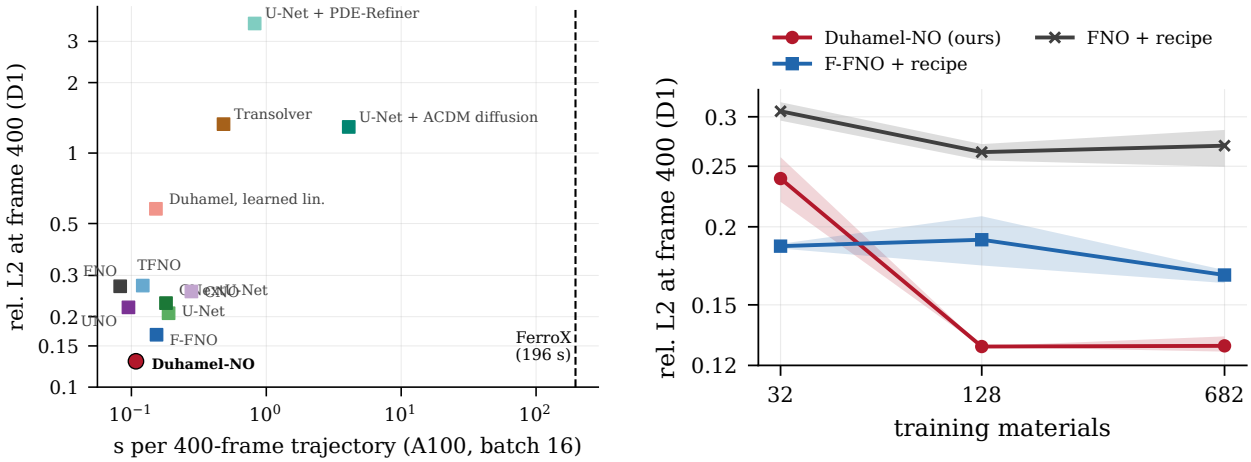


Figure 12: Left: error at frame 400 against wall-clock cost per 400-frame trajectory (A100, batch 16), with the FerroX cost. **Right:** error against the number of training materials (the Fourier SSM is rebuilt from the same subset). With 128 materials Duhamel-NO is unchanged (0.129); with 32 it degrades to 0.239, where F-FNO is better (0.186).

E.5 Rollouts of every model

Figures 13 and 14 show one D1 held-out and one D2 trajectory for every model, from the same unseen noise. Each is the trajectory with the median Duhamel-NO error among the saved examples.

F The released benchmark

We release the three ferroelectric datasets of Table 3 as an open benchmark under the CC BY 4.0 license. The data are hosted on Hugging Face (doi:10.57967/hf/11169), and the code repository links to them. Each dataset is a folder of one file per trajectory and a metadata table (CSV) with one row per trajectory, giving the Landau coefficients (α, β, γ), the derived coordinates (P_S, r), the solver time step and frame interval, the noise seed and the split used in this paper.

- **D1** (1001 trajectories) and **D2** (300) are arrays of shape $401 \times 10 \times 64 \times 64$ (frames, depth layers, in-plane grid), float32, in C/m^2. D1 carries the splits train, validation, held-out interior and held-out edge; D2 carries its region (A, B_{units}, B_α, C, D).

- **Hysteresis** (174 loops) holds the field at every saved solver step, with the step index and the applied voltage, and the material, ε_{de}, V_{max}, seed and split (train, validation, new noise, held-out material, closed set).

The ferroelectric film occupies depth layers 8–17 of the simulated stack; the eight dielectric layers carry no polarization (they are exactly zero in every saved frame) and are omitted. A small Python module loads a trajectory and computes the pattern correlation and relative error of Section 6.2, so that results on the benchmark are comparable with Table 1. The FerroX input files and the job lists reproduce every trajectory from its seed.

G Voltage-driven hysteresis

Setup. The same capacitor, with $\varepsilon_{de} = 100$ (with 10 the loop is too narrow to resolve), is driven by a quasi-static voltage sweep $0 \to +V_{max} \to -V_{max} \to +V_{max}$ from the unpoled random field. The top electrode potential is stepped in 0.125 V increments, each held until the electrostatic solve converges, with $V_{max} = 1.0$, 1.5 or 2.0 V and frames every 20 solver steps; a loop takes 440–2765 frames. Every model is retrained on 90 closed loops and rolls out the whole sweep from the first frame, with the voltage as an input. The loop error is the mean absolute deviation of the polarization–voltage curve, in units of P_S.

Results. CNO is the best model on held-out materials (0.078; Table 9). Duhamel-NO is second (0.138), with 1.76× the loop error of CNO, although its first-frame error is 4.34× lower than that of the best other model. Its Fourier SSM is measured around the unpoled state, whereas during a sweep the film spends most of its time near saturation, where the linearization is different; and at 20 solver steps per frame only part of the wavenumber bins, not including the uniform mode that carries the switched polarization, can be fitted. Voltage-driven switching is therefore the task on which Duhamel-NO is least competitive. We test on 30 loops of held-out materials, on new noise for training materials, and on an independent set of ten closed loops. Figure 15 shows the deviation of the predicted polarization along the sweep, and Figure 16 the predicted loops of held-out materials.

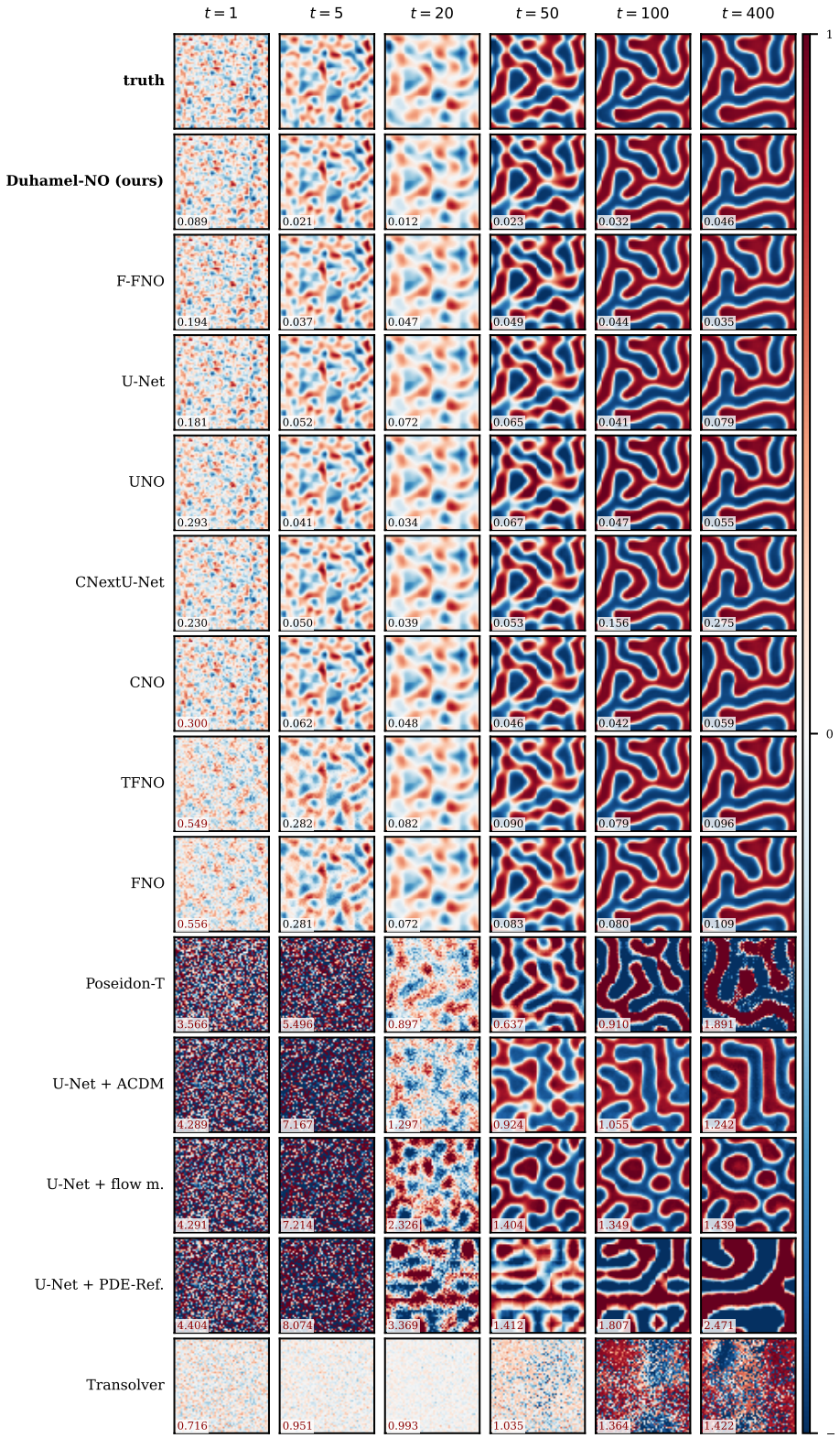


Figure 13: D1 held-out material, every model (top ferroelectric layer; field divided by the maximum of the truth at that frame; numbers are relative errors of the frame, red above 0.3).

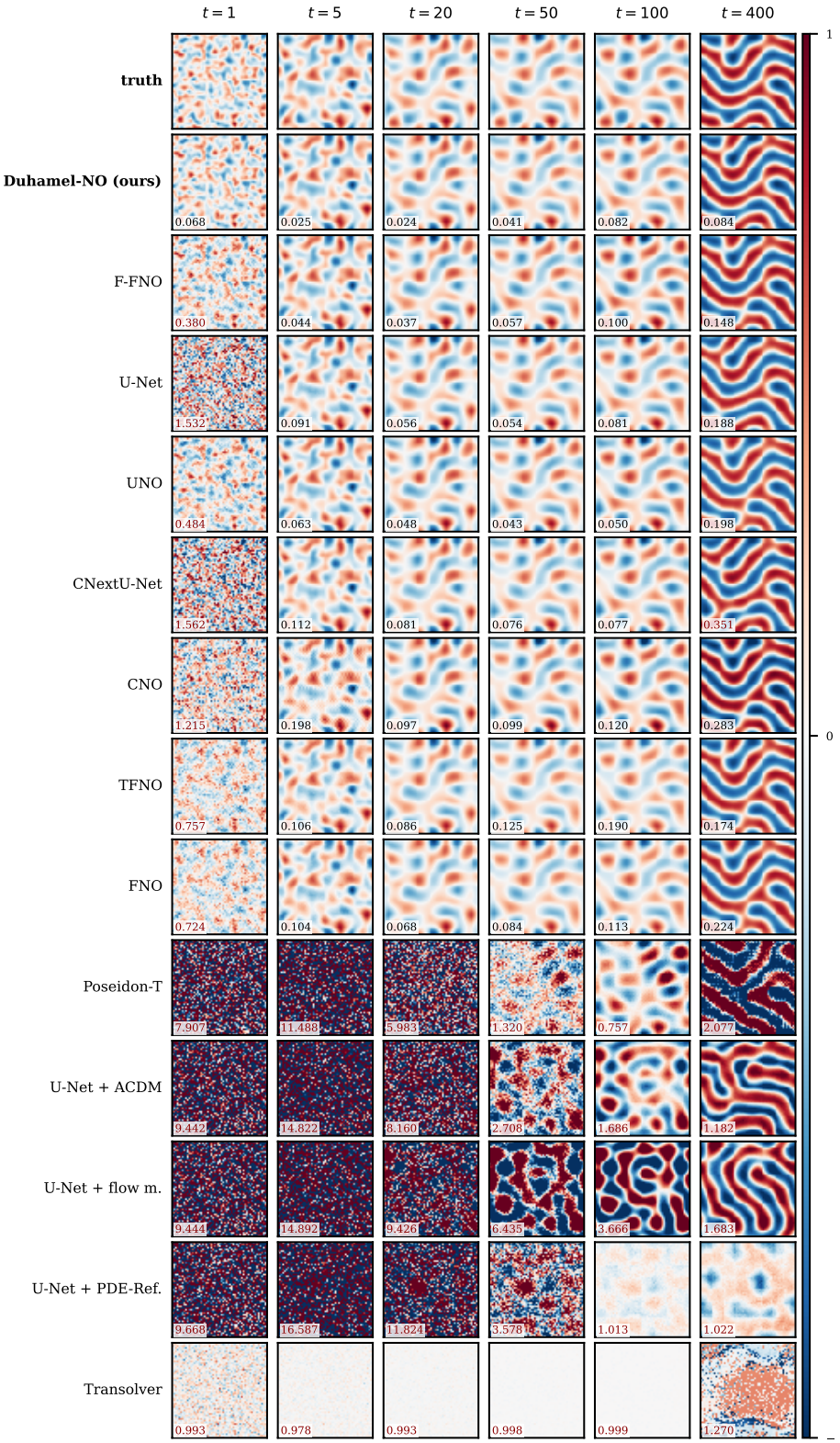


Figure 14: D2 material (out of range, no retraining), every model, as Figure 13.

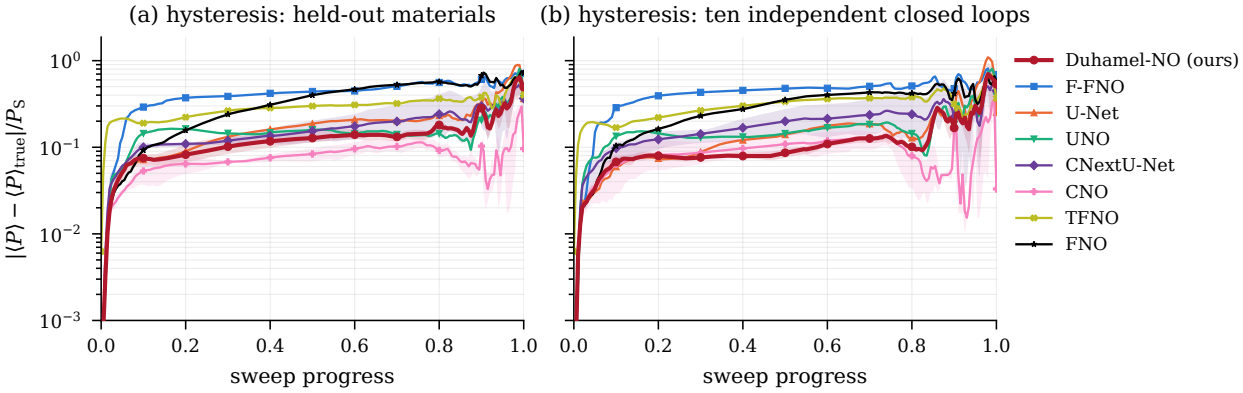


Figure 15: Hysteresis error along the sweep. Deviation of the mean polarization from FerroX, in units of P_S, on held-out materials (a) and on ten independent closed loops (b); its time mean is the loop error. Mean over loops, and bands show the range over training seeds.

Table 9: Hysteresis, closed loops: loop error (mean ± std over seeds) on held-out materials, on new noise for training materials, and on an independent set of ten closed loops; field correlation on held-out materials.

model	seeds	held-out material	new noise	extra closed set	field corr. (held-out mat.)
CNO	3	0.078±0.004	0.094±0.012	0.092±0.019	0.722±0.007
Duhamel-NO (ours)	3	0.138±0.014	0.128±0.003	0.121±0.012	0.617±0.042
UNO	1	0.164	0.159	0.172	0.189
CNextU-Net	3	0.171±0.052	0.184±0.055	0.191±0.064	0.455±0.120
U-Net	1	0.194	0.183	0.165	0.174
TFNO	1	0.295	0.310	0.309	0.074
FNO	1	0.361	0.245	0.310	0.397
F-FNO	1	0.431	0.409	0.439	0.264

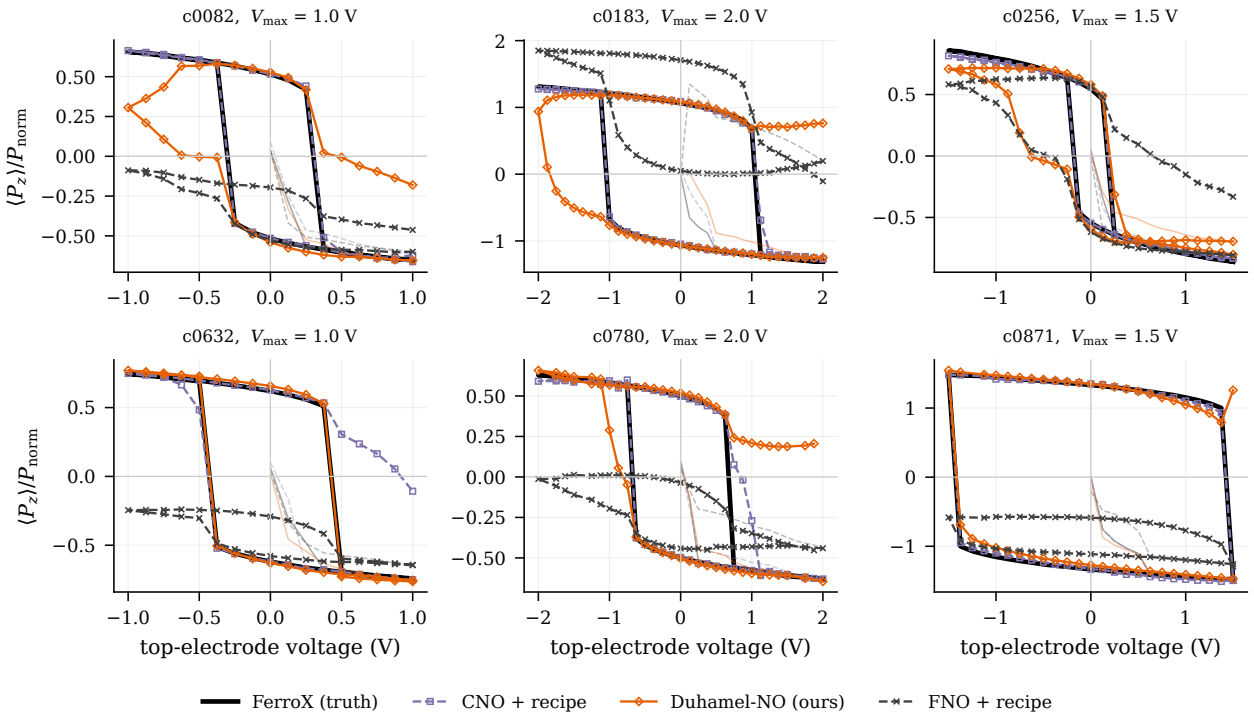


Figure 16: Closed polarization–voltage loops of held-out materials, predicted by rollout of the whole sweep from the unpoled random field (virgin branch drawn lightly).

H Allen–Cahn

Data. We use the 2D Allen–Cahn equation $\partial_t u = -M(au + bu^3 - e\nabla^2 u)$ with $a < 0$, which is the single-layer limit of TDGL without depolarization or a sextic term and shares its noise-seeded structure. We generated these data ourselves on a periodic 64^2 grid from small random noise, with an exponential-integrator solver. Such a solver favours an exponential-Euler model such as Duhamel-NO.

Setup. Each split has 900 training, 100 validation and 128 test trajectories. The amplitude unit $U = \sqrt{|a|/b}$, the wall width $\mu = \sqrt{e/|a|}$ and the grid spacing in wall widths $\Pi_h = \Delta x/\mu$ are sampled for training ($U \in [0.5, 2]$, $|a| \in [0.5, 5]$, $\Pi_h \in [0.30, 0.70]$). Two test splits leave this range, *new units* ($U \in [4, 10]$, $|a| \in [20, 100]$, same Π_h) and *new wall widths* ($\Pi_h \in [0.85, 1.20]$). The in-range test materials are also simulated on 32^2 and 128^2 grids of the same domain, and every model trained on 64^2 is rolled out on them unchanged. Table 10 gives the final-frame errors on every split and grid.

Results. On Allen–Cahn, Duhamel-NO learns its linear symbol as a polynomial in the dimensionless wavenumber $k\mu$, which does not depend on the grid. A learned symbol suffices here, unlike in TDGL (Table 2), because there are no depth layers to couple. In range, its final error is 0.012 (F-FNO 0.043). On new wall widths it reaches 0.075, which is 3.5× lower than the best baseline (0.265, TFNO).

Table 10: Allen–Cahn: relative L_2 error at the final frame (mean over seeds; **best**, second). New units have amplitude and time units outside the training range. New wall widths have the wall width relative to the grid outside it. The zero-shot grids hold the in-range test materials on 32^2 and 128^2, with models trained on 64^2.

	trained grid (64^2)			zero-shot grid (in range)	
model	in range	new units	new wall widths	32^2	128^2
Duhamel-NO (ours)	**0.012**	**0.010**	**0.075**	0.124	**0.010**
F-FNO	0.043	0.042	0.416	**0.049**	0.086
CNextU-Net	0.057	0.051	1.023	0.879	0.909
U-Net	0.088	0.066	0.378	0.857	0.854
CNO	0.097	0.090	0.667	0.884	0.859
UNO	0.092	0.090	0.920	0.717	0.194
TFNO	0.120	0.102	0.265	0.094	0.271
FNO	0.150	0.148	0.443	0.115	0.276